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(54) **REMOTE WEB BASED CONTROLS FOR AUTONOMOUS OPERATIONS**

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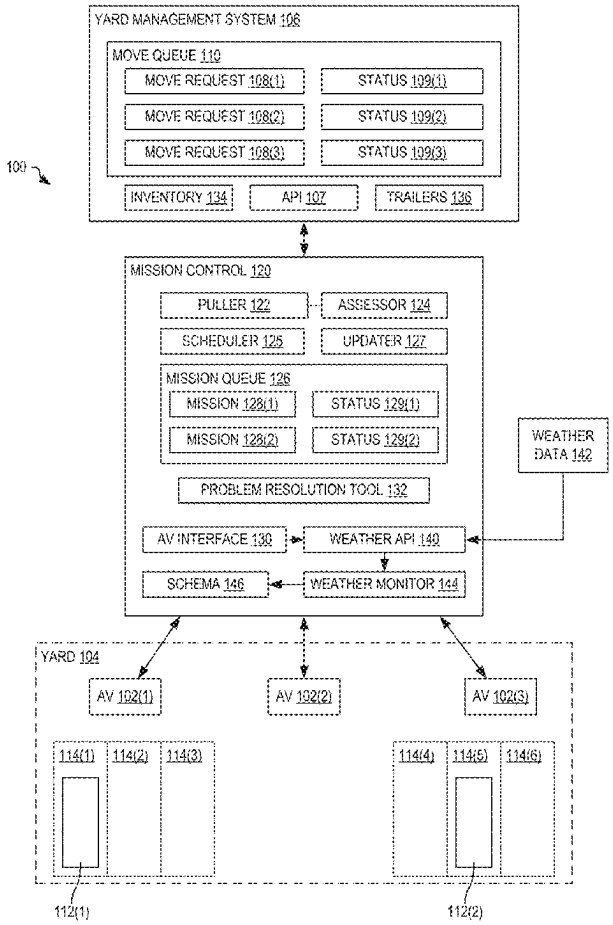
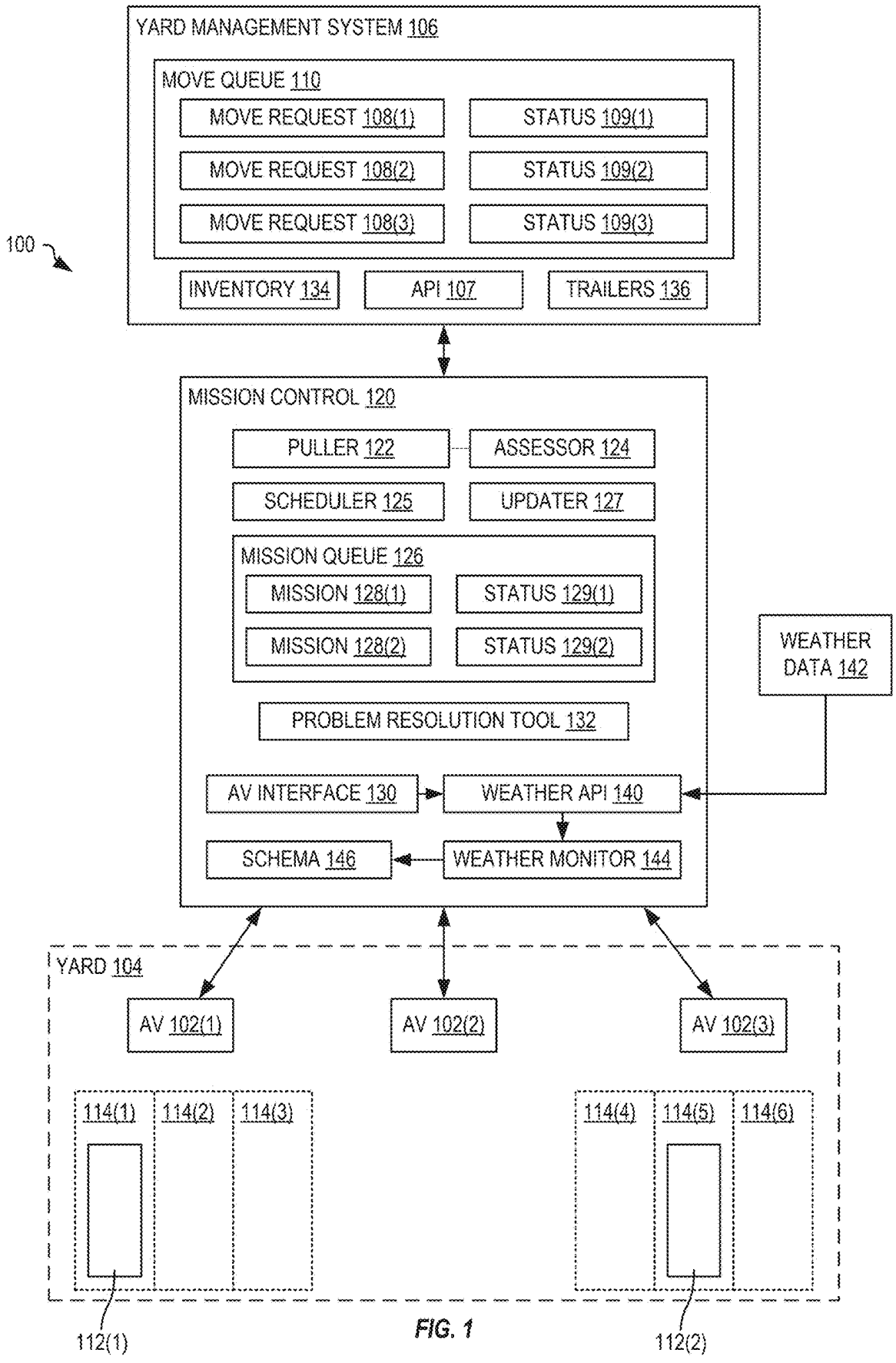
(57) **ABSTRACT**

A mission control may receive, from a yard management system (YMS), a move request identifying a destination spot in a yard and one or more of a trailer identifier of a trailer to be moved, a pick-up spot of the trailer, and a trailer type. Mission control determines whether the move request is feasible for an autonomous vehicle (AV). When the move request is feasible, mission control generates a mission defining directives to control the AV to move the trailer from the pick-up spot to the destination spot and sets, via an application programming interface (API) of the YMS, a status of the move request to indicate autonomous move is scheduled.

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MOVE QUEUE 110
MOVE REQUEST 108(1)
STATUS 109(1)
MOVE REQUEST 108(2)
STATUS 109(2)
MOVE REQUEST 108(3)
STATUS 109(3)
100
INVENTORY 134
API 107
TRAILERS 136
MISSION CONTROL 120
PULLER 122
ASSESSOR 124
SCHEDULER 125
UPDATER 127
MISSION QUEUE 126
MISSION 128(1)
STATUS 129(1)
MISSION 128(2)
STATUS 129(2)
WEATHER DATA 142
PROBLEM RESOLUTION TOOL 132
AV INTERFACE 130
WEATHER API 140
SCHEMA 146
WEATHER MONITOR 144
YARD 104
AV 102(1)
AV 102(2)
AV 102(3)
114(1)
114(2)
114(3)
114(4)
114(5)
114(6)
112(1)
112(2)

FIG. 1

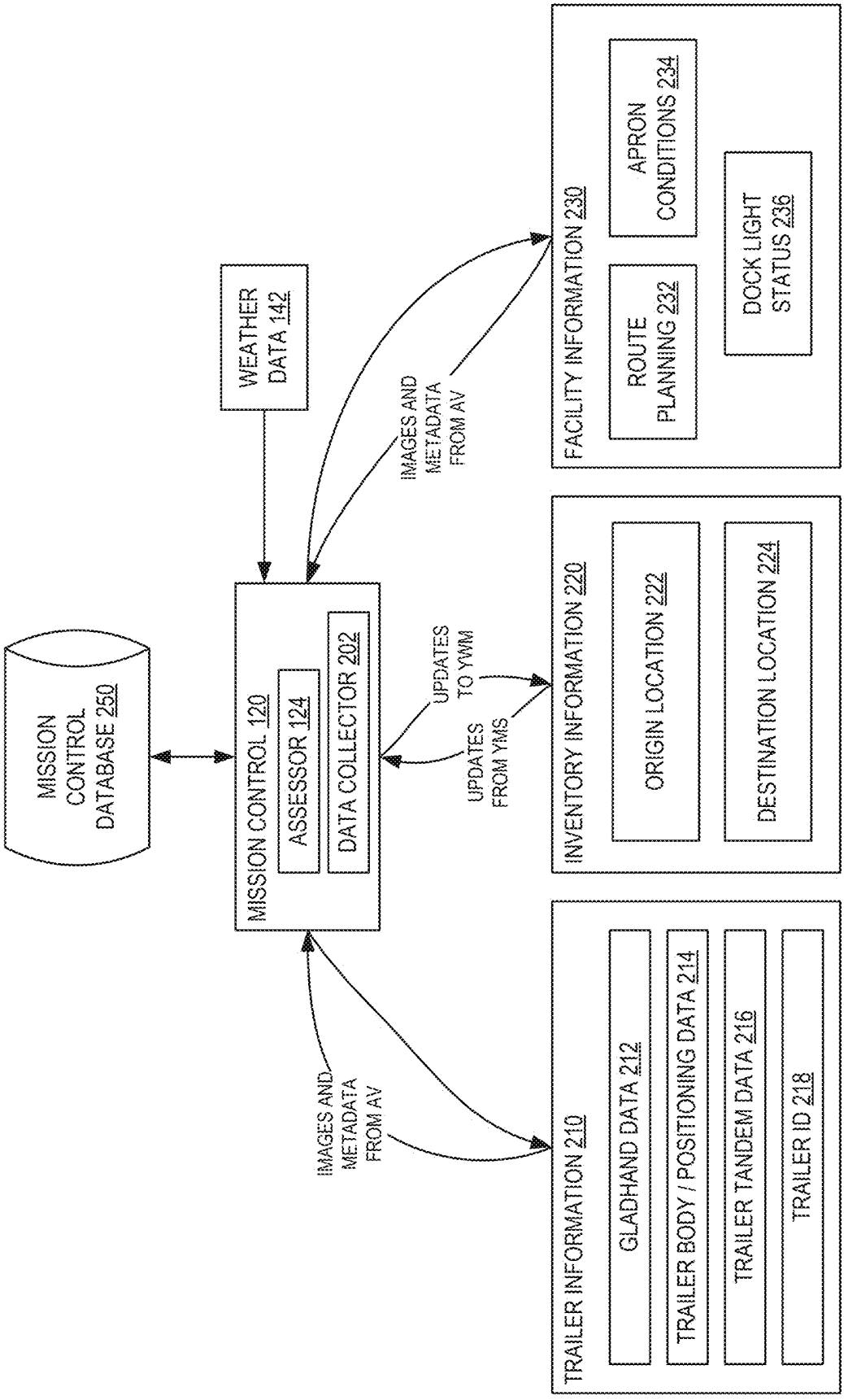
MISSION CONTROL DATABASE 250
MISSION CONTROL 120
ASSESSOR 124
DATA COLLECTOR 202
WEATHER DATA 142
IMAGES AND METADATA FROM AV
FACILITY INFORMATION 230
ROUTE PLANNING 232
APRON CONDITIONS 234
DOCK LIGHT STATUS 236
UPDATES TO YMS
UPDATES FROM YMS
INVENTORY INFORMATION 220
ORIGIN LOCATION 222
DESTINATION LOCATION 224
IMAGES AND METADATA FROM AV
TRAILER INFORMATION 210
GLADHAND DATA 212
TRAILER BODY / POSITIONING DATA 214
TRAILER TANDEM DATA 216
TRAILER ID 218

FIG. 2

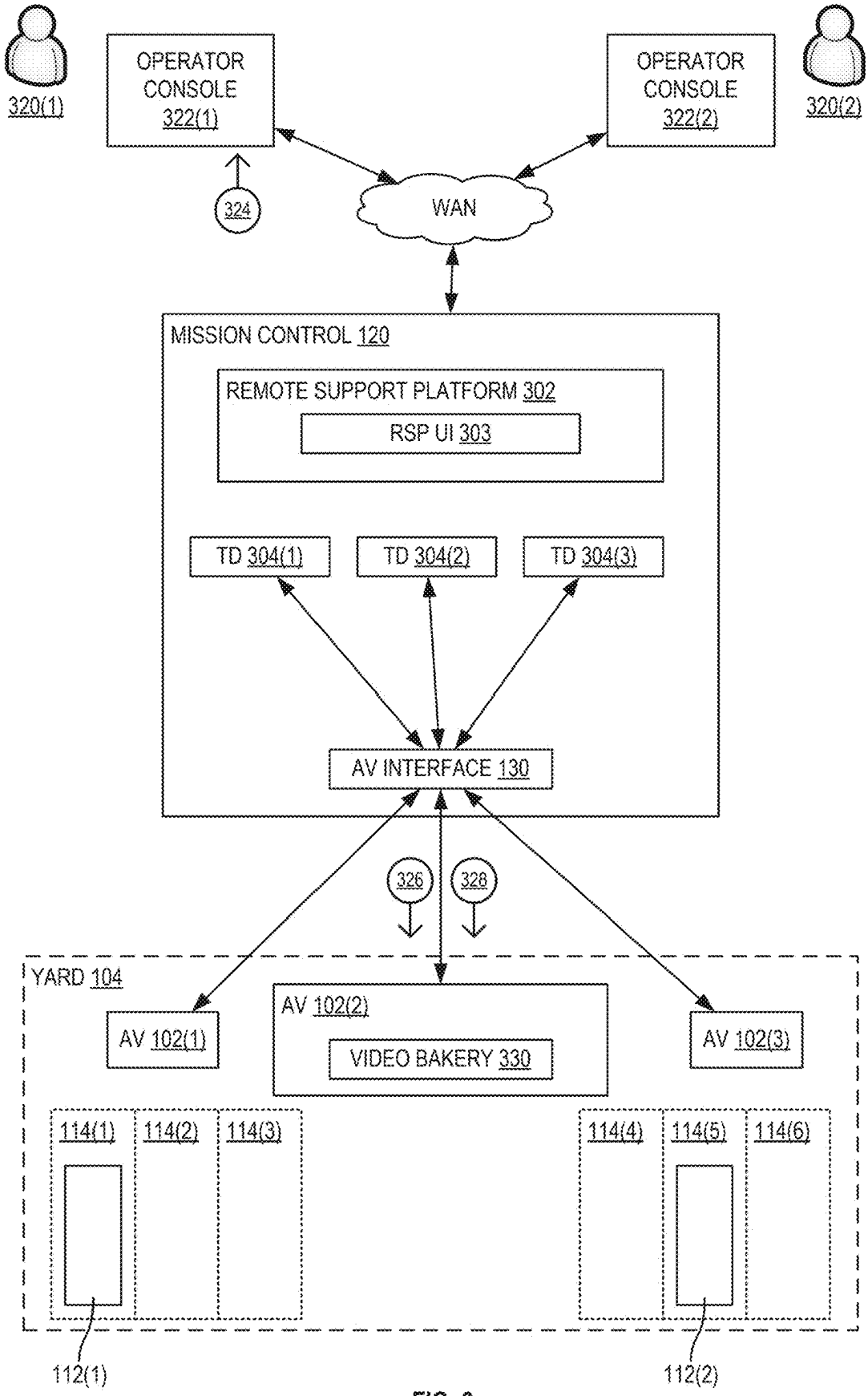
320(1)
OPERATOR CONSOLE 322(1)
OPERATOR CONSOLE 322(2)
320(2)
324
WAN
MISSION CONTROL 120
REMOTE SUPPORT PLATFORM 302
RSP UI 303
TD 304(1)
TD 304(2)
TD 304(3)
AV INTERFACE 130
326
328
YARD 104
AV 102(1)
AV 102(2)
VIDEO BAKERY 330
AV 102(3)
114(1)
114(2)
114(3)
114(4)
114(5)
114(6)
112(1)
112(2)

FIG. 3

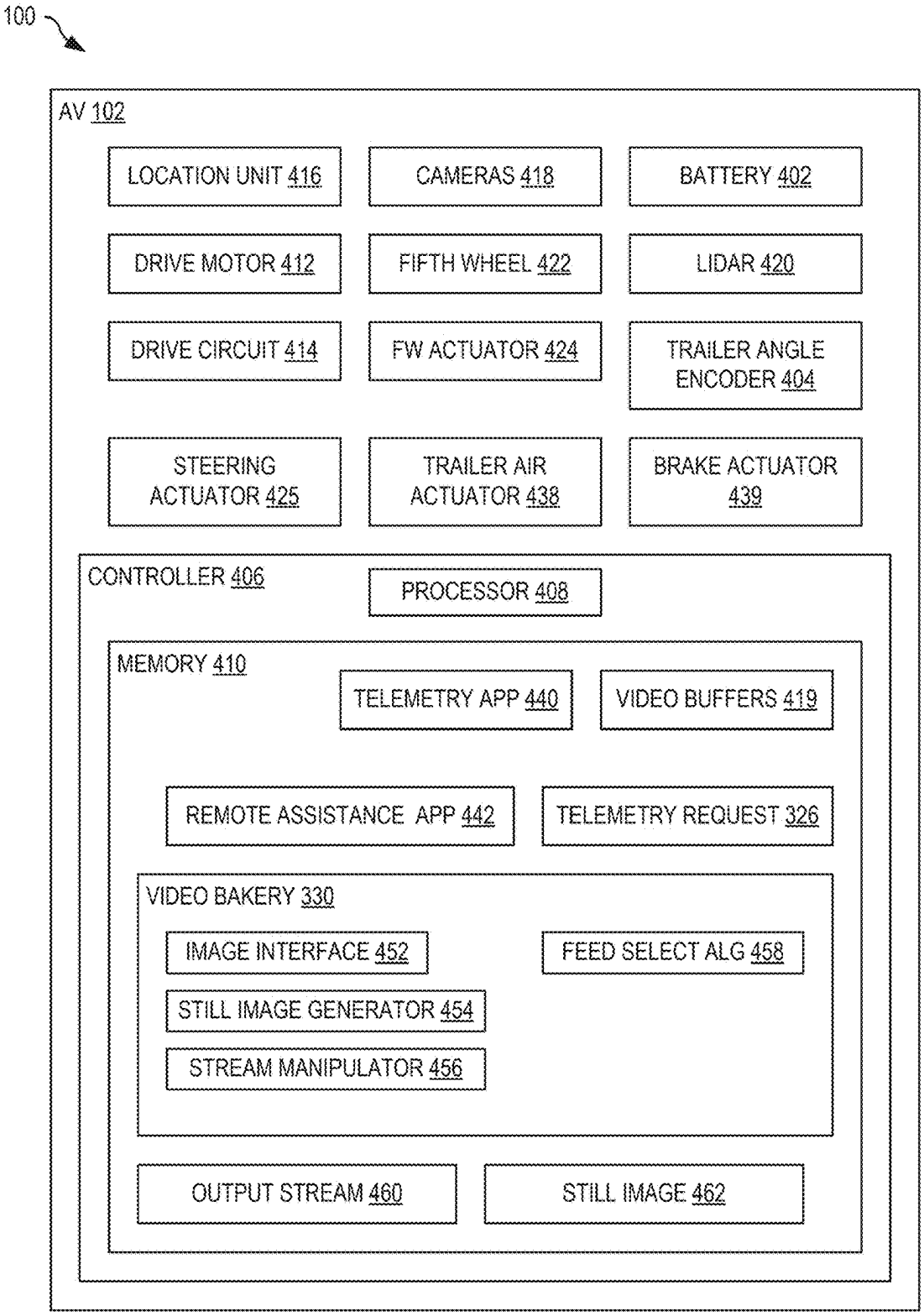
100
AV 102
LOCATION UNIT 416
CAMERAS 418
BATTERY 402
DRIVE MOTOR 412
FIFTH WHEEL 422
LIDAR 420
DRIVE CIRCUIT 414
FW ACTUATOR 424
TRAILER ANGLE ENCODER 404
STEERING ACTUATOR 425
TRAILER AIR ACTUATOR 438
BRAKE ACTUATOR 439
CONTROLLER 406
PROCESSOR 408
MEMORY 410
TELEMETRY APP 440
VIDEO BUFFERS 419
REMOTE ASSISTANCE APP 442
TELEMETRY REQUEST 326
VIDEO BAKERY 330
IMAGE INTERFACE 452
FEED SELECT ALG 458
STILL IMAGE GENERATOR 454
STREAM MANIPULATOR 456
OUTPUT STREAM 460
STILL IMAGE 462

FIG. 4

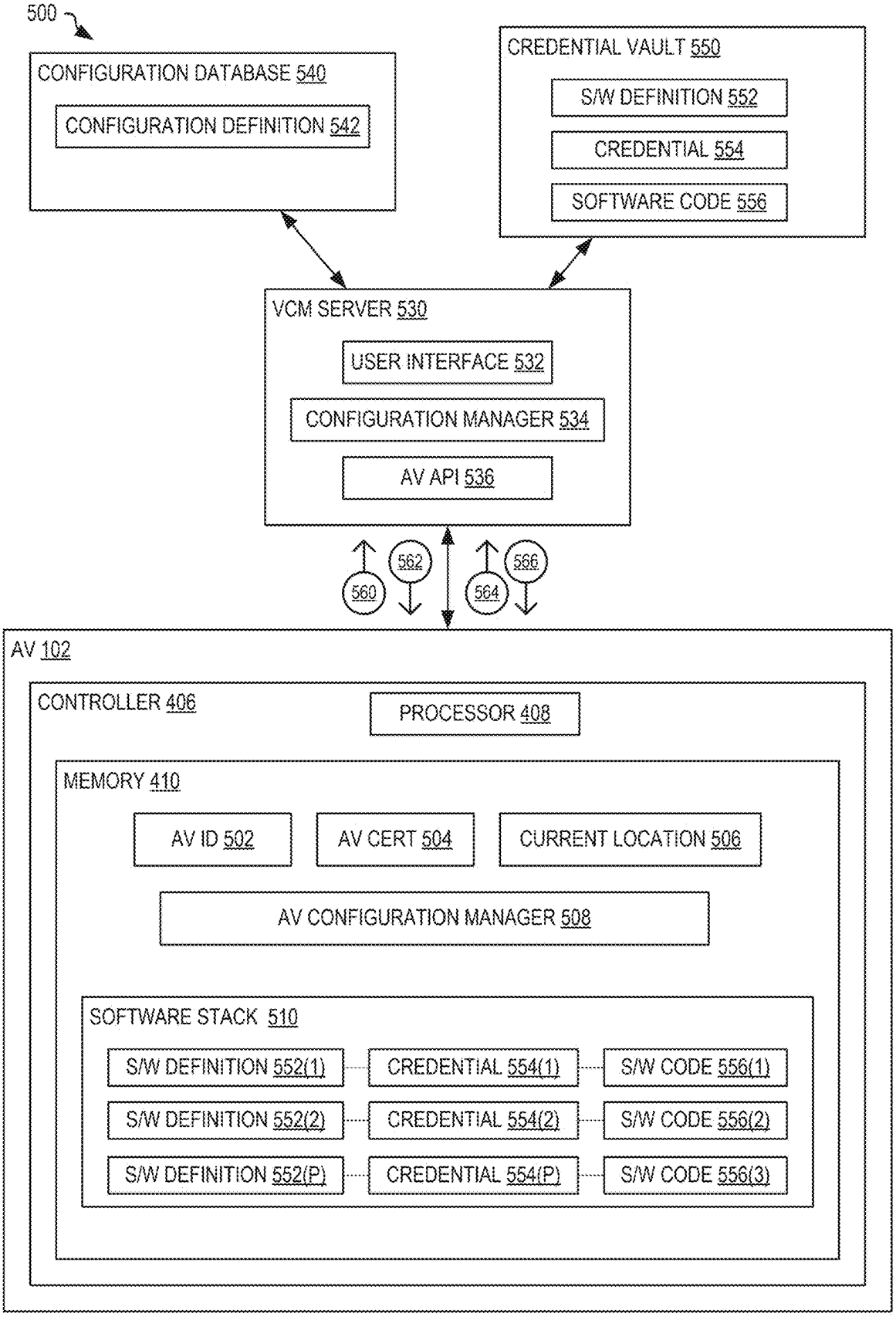
500
CONFIGURATION DATABASE 540
CONFIGURATION DEFINITION 542
CREDENTIAL VAULT 550
S/W DEFINITION 552
CREDENTIAL 554
SOFTWARE CODE 556
VCM SERVER 530
USER INTERFACE 532
CONFIGURATION MANAGER 534
AV API 536
560
562
564
566
AV 102
CONTROLLER 406
PROCESSOR 408
MEMORY 410
AV ID 502
AV CERT 504
CURRENT LOCATION 506
AV CONFIGURATION MANAGER 508
SOFTWARE STACK 510
S/W DEFINITION 552(1)
CREDENTIAL 554(1)
S/W CODE 556(1)
S/W DEFINITION 552(2)
CREDENTIAL 554(2)
S/W CODE 556(2)
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FIG. 5

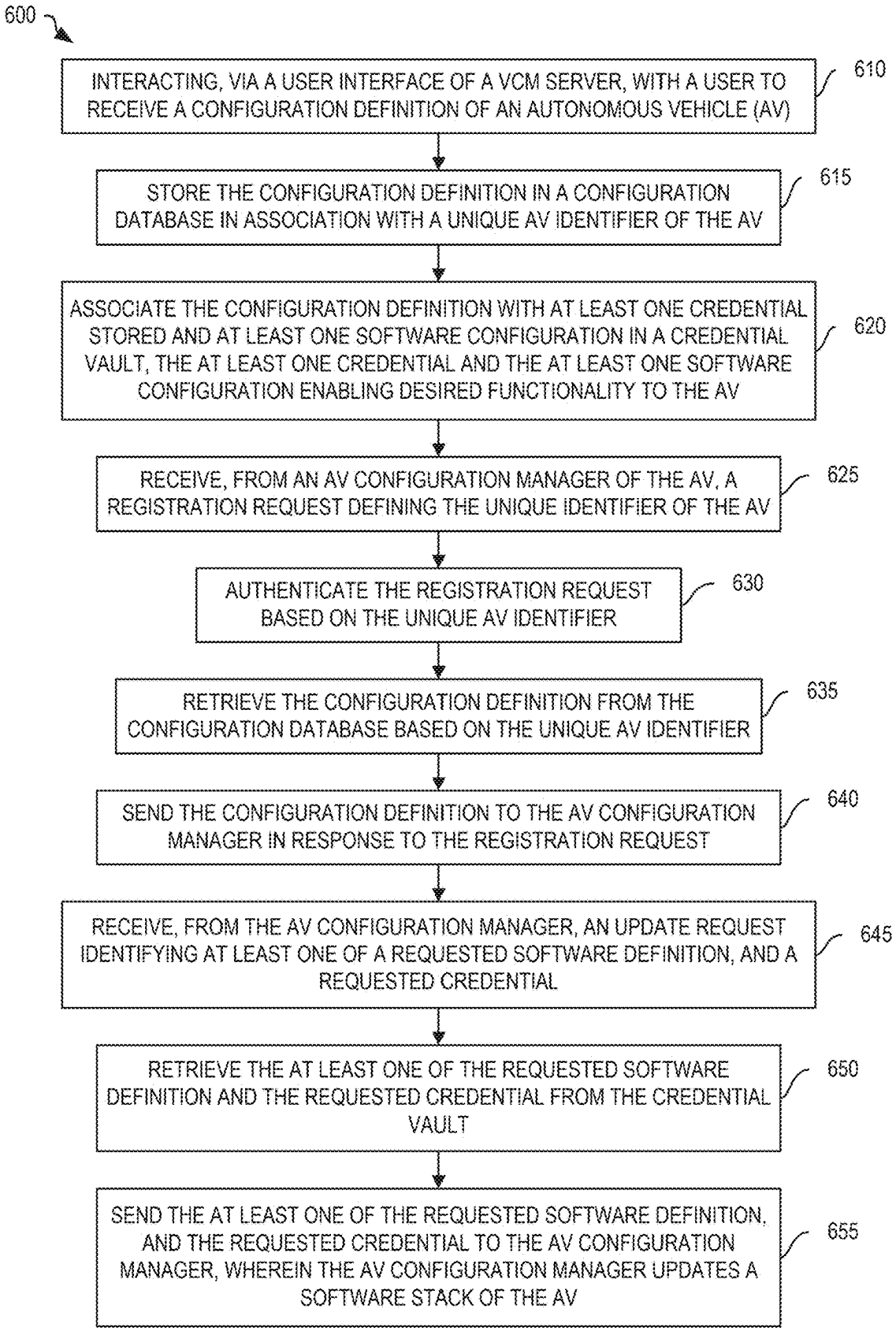
600
INTERACTING, VIA A USER INTERFACE OF A VCM SERVER, WITH A USER TO RECEIVE A CONFIGURATION DEFINITION OF AN AUTONOMOUS VEHICLE (AV)
610
STORE THE CONFIGURATION DEFINITION IN A CONFIGURATION DATABASE IN ASSOCIATION WITH A UNIQUE AV IDENTIFIER OF THE AV
615
ASSOCIATE THE CONFIGURATION DEFINITION WITH AT LEAST ONE CREDENTIAL STORED AND AT LEAST ONE SOFTWARE CONFIGURATION IN A CREDENTIAL VAULT, THE AT LEAST ONE CREDENTIAL AND THE AT LEAST ONE SOFTWARE CONFIGURATION ENABLING DESIRED FUNCTIONALITY TO THE AV
620
RECEIVE, FROM AN AV CONFIGURATION MANAGER OF THE AV, A REGISTRATION REQUEST DEFINING THE UNIQUE IDENTIFIER OF THE AV
625
AUTHENTICATE THE REGISTRATION REQUEST BASED ON THE UNIQUE AV IDENTIFIER
630
RETRIEVE THE CONFIGURATION DEFINITION FROM THE CONFIGURATION DATABASE BASED ON THE UNIQUE AV IDENTIFIER
635
SEND THE CONFIGURATION DEFINITION TO THE AV CONFIGURATION MANAGER IN RESPONSE TO THE REGISTRATION REQUEST
640
RECEIVE, FROM THE AV CONFIGURATION MANAGER, AN UPDATE REQUEST IDENTIFYING AT LEAST ONE OF A REQUESTED SOFTWARE DEFINITION, AND A REQUESTED CREDENTIAL
645
RETRIEVE THE AT LEAST ONE OF THE REQUESTED SOFTWARE DEFINITION AND THE REQUESTED CREDENTIAL FROM THE CREDENTIAL VAULT
650
SEND THE AT LEAST ONE OF THE REQUESTED SOFTWARE DEFINITION, AND THE REQUESTED CREDENTIAL TO THE AV CONFIGURATION MANAGER, WHEREIN THE AV CONFIGURATION MANAGER UPDATES A SOFTWARE STACK OF THE AV
655

FIG. 6

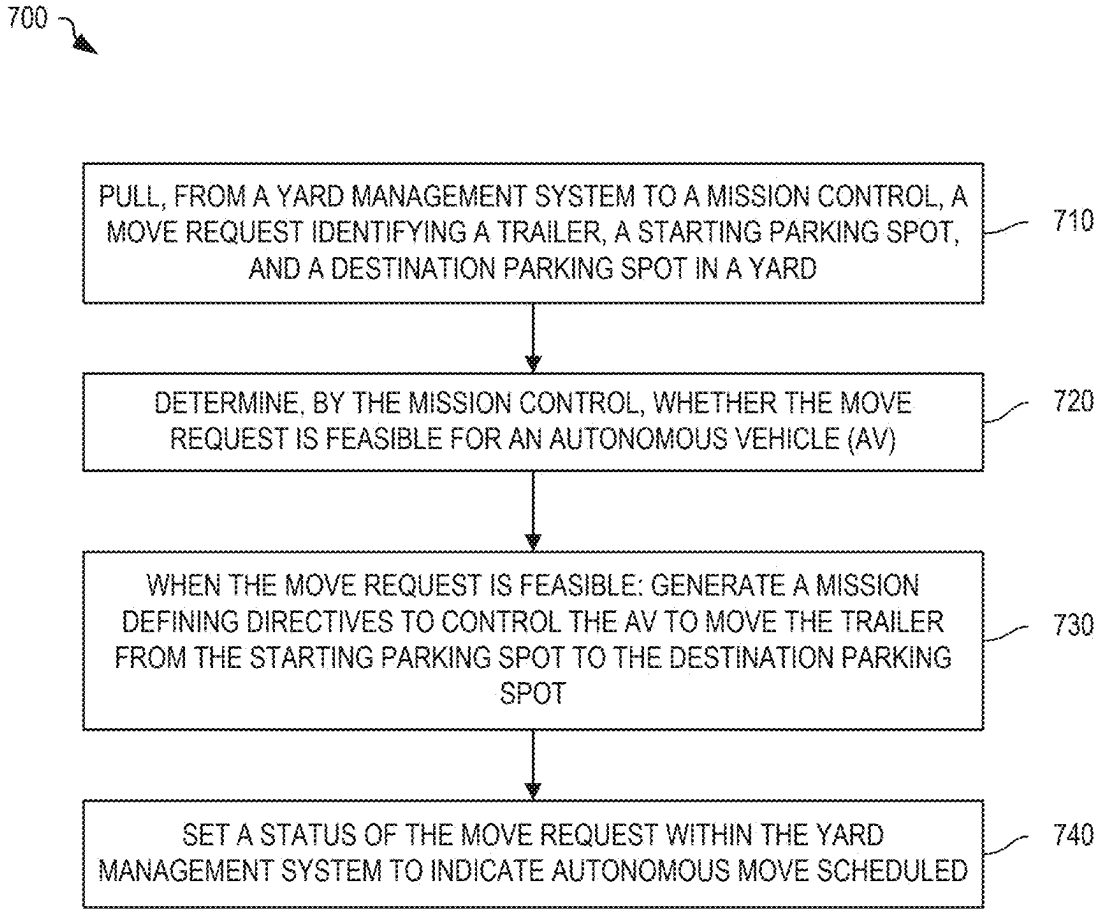
700
PULL, FROM A YARD MANAGEMENT SYSTEM TO A MISSION CONTROL, A MOVE REQUEST IDENTIFYING A TRAILER, A STARTING PARKING SPOT, AND A DESTINATION PARKING SPOT IN A YARD
710
DETERMINE, BY THE MISSION CONTROL, WHETHER THE MOVE REQUEST IS FEASIBLE FOR AN AUTONOMOUS VEHICLE (AV)
720
WHEN THE MOVE REQUEST IS FEASIBLE: GENERATE A MISSION DEFINING DIRECTIVES TO CONTROL THE AV TO MOVE THE TRAILER FROM THE STARTING PARKING SPOT TO THE DESTINATION PARKING SPOT
730
SET A STATUS OF THE MOVE REQUEST WITHIN THE YARD MANAGEMENT SYSTEM TO INDICATE AUTONOMOUS MOVE SCHEDULED
740

FIG. 7

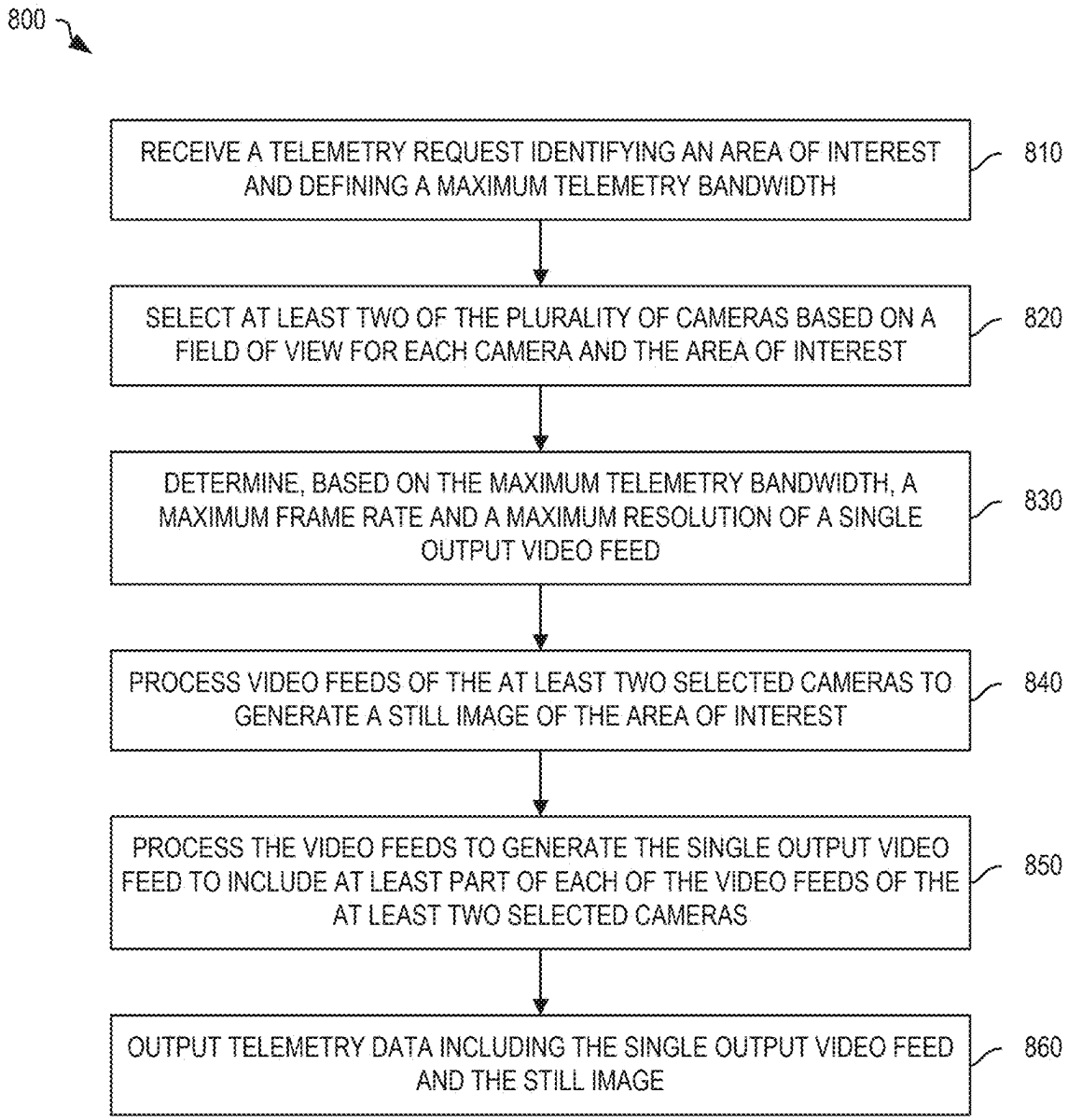
800
RECEIVE A TELEMETRY REQUEST IDENTIFYING AN AREA OF INTEREST AND DEFINING A MAXIMUM TELEMETRY BANDWIDTH
810
SELECT AT LEAST TWO OF THE PLURALITY OF CAMERAS BASED ON A FIELD OF VIEW FOR EACH CAMERA AND THE AREA OF INTEREST
820
DETERMINE, BASED ON THE MAXIMUM TELEMETRY BANDWIDTH, A MAXIMUM FRAME RATE AND A MAXIMUM RESOLUTION OF A SINGLE OUTPUT VIDEO FEED
830
PROCESS VIDEO FEEDS OF THE AT LEAST TWO SELECTED CAMERAS TO GENERATE A STILL IMAGE OF THE AREA OF INTEREST
840
PROCESS THE VIDEO FEEDS TO GENERATE THE SINGLE OUTPUT VIDEO FEED TO INCLUDE AT LEAST PART OF EACH OF THE VIDEO FEEDS OF THE AT LEAST TWO SELECTED CAMERAS
850
OUTPUT TELEMETRY DATA INCLUDING THE SINGLE OUTPUT VIDEO FEED AND THE STILL IMAGE
860

FIG. 8

REMOTE WEB BASED CONTROLS FOR AUTONOMOUS OPERATIONS

RELATED APPLICATION

[0001] This application claims priority to U.S. Patent Application No. 63/558,476, titled "Remote Web Based Controls for Autonomous Operations," filed Feb. 27, 2024, and which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] A yard management system (YMS) controls a customer's inventory of trailers within a yard (e.g., a warehouse with loading docks and a staging area) and schedules movement of the trailers between parking spots and loading docks for loading and unloading of trailer content. The YMS maintains a queue of trailer move requests, some being handled by manually driven vehicles, and others being sent to a mission control for handling by autonomous vehicles. When an autonomous vehicle is unable to autonomously complete the move request, mission control notifies the YMS that the move cannot be completed autonomously, which causes the YMS to reschedule the move for manual handling. Disadvantageously, this causes disruption to the scheduled manual movements, resulting in inefficiency within the yard. Further, status of the autonomous vehicles is not visible within the YMS, and thus the user of the YMS is unable to see a true status of all moves within the yard. This requires the user to simultaneously monitor information from mission control to be aware of all trailer movement activity within the yard, which is inconvenient and inefficient.

[0003] These problems also apply to other application areas where autonomous vehicles work alongside yard trucks, such as in ports, intermodal rail terminals, and any other location where trailers are moved autonomously such as where semi-trailers are used for highway freight movement and shipping containers are moved on wheeled-chassis. The problem also applies to use of Terminal Operating Systems (TOS), Warehouse Management Systems (WMS), and Transportation Management Systems (TMS) that are used for tracking movement of the trailers and containers.

SUMMARY

[0004] One aspect of the present embodiments includes the realization that it is difficult to monitor both (a) a yard management system (YMS) that defines asset movements and tracks manually driven vehicles, and (b) a mission control that provides management of autonomous vehicles within the yard, to track all vehicle and inventory movements. The present embodiments solve this problem by interfacing mission control directly with the YMS (via APIs) and allowing mission control to update the status of the autonomous trailer movements directly within the YMS. Advantageously, this allows a user of the YMS to see the status of all trailer moves (manual and autonomous) within the yard.

[0005] Another aspect of the present embodiments includes the realization that it is inefficient for the user of the YMS to send certain ones of the trailer move requests to mission control for autonomous handling without monitoring the status of mission control. The present embodiments solve this problem by allowing mission control to automatically select ones of the move requests stored on the YMS that could be handled by autonomous vehicles, and updating the status of the selected moves within the YMS. Advantageously, mission control selects move requests based on certain criteria that allow for efficient operation of the autonomous vehicles, leaving others to be handled manually. Further, mission control monitors (grooms) the yard during normal operation and automatically updates the status of parking spots and loading docks in an inventory of the YMS, thereby making the information displayed to the user of the YMS more current and thereby increasing yard efficiencies. For example, AVs operating within the yard collect information of parking spots that are nearby and/or passed. Using the collected information, mission control determines when a trailer is positioned in a parking spot, which parking spots are empty, and automatically updates the status of the relevant parking spot in the YMS, thereby correcting the YMS inventory for trailers that are parked incorrectly.

[0006] Another aspect of the present embodiments includes the realization that mission control may determine that a destination spot for a move is not empty. The present embodiments solve this problem by having mission control find a nearby empty spot within the YMS inventory and generate an updated mission plan to direct an AV to park the trailer in the nearby empty spot instead of the destination spot.

[0007] Another aspect of the present embodiments includes the realization that an AV may determine that a destination spot for a move is not empty. The AV uses imagery and deep learning algorithms to determine when a destination spot is empty, for example. The present embodiments solve this problem by having the AV request an alternative empty nearby spot from mission control. Mission control generates a updated mission plan to use the alternative empty nearby spot when found in the YMS inventory, and sends the updated mission plan to the AV, causing the AV to park the trailer in the alternative empty nearby spot. When mission control cannot find an empty nearby spot, mission control directs the AV to search nearby for an empty spot. For example, mission control may direct the AV to traverse the apron looking for empty spots. When the AV finds an empty spot, the AV identifies the empty spot to mission control and receives an updated mission plan to deposit the trailer in the empty spot.

[0008] Another aspect of the present embodiments includes the realization that when a move is received to move a specific trailer to a destination spot, mission control searches a yard inventory for the specific trailer to identify a pick-up spot, and generates a mission plan to cause an AV to move the specific trailer from the pick-up spot to the destination spot. When the AV identifies the specific trailer in a different spot while moving towards the requested pick up spot, the AV communicates the different pick-up spot to mission control and awaits an updated mission to mode the specific trailer from the different spot to the destination spot. When, as the AV moves towards the pick-up spot, mission control receives updated inventory information defining a new spot for the specific trailer from another AV at the yard, mission control generates an updated mission for the AV that redirects the AV to the new spot. When the AV arrives at the pick-up spot, the AV uses imagery and deep learning algorithms to determine whether the requested trailer is at the pick-up spot. When the requested trailer is not at the pick-up spot, the AV requests a new pick-up spot for the requested trailer from mission control. When mission control deter-

mines a new pick-up spot for the specific trailer from the YMS inventory, mission control generates an updated mission plan to cause the AV to collect the specific trailer from the new pick up spot. Mission control send the updated mission plan to the AV, which causes the AV to proceed to the new pick up spot. When mission control does not determine a new pick-up spot for the requested trailer, mission control instructs the AV to search nearby for the specific trailer. For example, mission control may direct the AV to traverse the apron looking for nearby trailers that match the specific trailer. When the specific trailer is found by the AV, the AV identifies a new pick up spot to mission control and receives an updated mission plan to move the specific trailer from the new pick-up spot to the destination spot.

[0009] Another aspect of the present embodiments includes the realization that a move request may only identify a trailer type to be moved to a destination spot. For example, a move request may state "move an empty **28'** dry van to a destination spot." Advantageously, the YMS stores a status and location of trailers within a trailer database and the inventory. Accordingly, mission control searches for the trailer type in the yard inventory, and when a trailer matches the trailer type, mission control assigns the matching trailer to the move request and generates a mission to cause the AV to move the matching trailer from a pick-up spot (e.g., the current location of the matching trailer) to the destination location. The AV uses imagery and deep learning algorithms to identify the trailer at the pick-up spot, allowing mission control to verify it is of the correct type. When the identified trailer does not match the trailer type, the AV requests a new match for the trailer type from mission control. When mission control finds another trailer matching the requested trailer type in the inventory, mission control updates the mission plan with a new trailer identifier and a new pick-up spot. The updated mission is sent to the AV and causes the AV to proceed to the new pick up spot. When mission control does not have a trailer of requested trailer type in inventory, mission control instructs the AV to search nearby for at least one parked trailer and to detect its trailer identifier. For example, mission control may direct the AV to traverse the apron looking for a nearby trailer of the requested trailer type. If a trailer is found by the AV, the AV sends its identity and a new pick-up spot of the found trailer to mission control, which determines whether the identified trailer matches the trailer type. When a match is found, mission control generates a new mission to move the found trailer from the new pick-up spot to the destination.

[0010] Another aspect of the present embodiments includes the realization that certain move requests cannot be performed by an autonomous vehicle and returning these moves for manual handling disrupts the scheduling of drivers. The present embodiments solve this problem by determining, as early as possible and preferably prior to attempting the move by an autonomous vehicle, ones of the move requests listed in the YMS that cannot be handled autonomously. Advantageously, move requests flagged early as being non-autonomous cause less disruption to the drive schedule.

[0011] Another aspect of the present embodiments includes the realization that an autonomous vehicle occasionally needs manual assistance and that to allow an operator to assist an AV remotely requires a current, stable, and continuous display of conditions at the autonomous vehicle on an operator console. Where the assisting operator is remote from the yard, telemetering all information captured by the autonomous vehicle via the remote support platform to the operator console is not possible due to a limited bandwidth between the autonomous vehicle and the remote operator console without causing undesirable delays in information displayed to the operator. The present embodiments solve this problem by using a video bakery at the autonomous vehicle to intelligently select information sent to the remote support platform based on the reason for the request for remote assistance. Advantageously, the video bakery automatically selected relevant portions of the available image (still and video) feeds from the multiple cameras of the AV, adjusting the resolution and/or frame rate as needed, to form telemetry data with needed information at the best available quality (e.g., best frame rate and resolution) that is within the available bandwidth.

[0012] Another aspect of the present embodiments includes the realization that manual software configuration updates of AVs deployed at a customer's yard is impractical, particularly where the updates apply to many AVs at many different locations, and where each AV requires a different configuration. The present embodiments solve this problem by providing a vehicle credential management (VCM) server in the cloud that allows each AV to automatically update itself based on a configuration definition in a configuration database. Advantageously, an operator defines or updates the configuration definition to indicate needed software and/or credentials in the configuration database, where the configuration definition is associated with unique AV identifiers of AVs that should be provisioned. In response to receiving an update request from the AV (e.g., triggered during initialization or booting of the AV controller), the VCM server provides the AV with the updated software configuration, including corresponding credentials, from a secure credential vault.

[0013] In certain embodiments, the techniques described herein relate to a remote based control method for autonomous operation, including: receiving, in a mission control from a yard management system (YMS), a move request defining a destination spot in a yard and one or more of a trailer identifier of a trailer to be moved, a pick-up spot of the trailer, and a trailer type; determining, at the mission control, whether the move request is feasible for an autonomous vehicle (AV); and when the move request is feasible: generating a mission defining directives to control the AV to move the trailer from the pick-up spot to the destination spot; and setting, via an application programming interface (API) of the YMS, a status of the move request to indicate autonomous move scheduled.

BRIEF DESCRIPTION OF THE FIGURES

[0014] FIG. **1** is a schematic diagram illustrating example remote web based controls for autonomous operation, in embodiments.

[0015] FIG. **2** is a block diagram illustrating operation of the assessor of FIG. **1** in further example detail.

[0016] FIG. **3** is a block diagram illustrating mission control of FIG. **1** with a remote support platform that allows a remote operator to use an operator console to remotely monitor and control any one AV, in embodiments.

[0017] FIG. **4** is a block diagram illustrating example detail of the AV of FIGS. **1** and **3**, in embodiments.

[0018] FIG. **5** shows one example VCM system that facilitates autonomous configuration of the AV of FIG. **1**, in embodiments.

[0019] FIG. **6** is a flowchart illustrating one example VCM method, in embodiments.

[0020] FIG. **7** is a flowchart illustrating one example remote based control method for autonomous operation, in embodiments.

[0021] FIG. **8** is a flowchart illustrating one example video bakery method for an autonomous vehicle having a plurality of cameras, in embodiments.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0022] The following embodiments and examples relate to a yard management system (YMS) that controls a customer's inventory of trailers within a yard (e.g., a warehouse with loading docks and a staging area) and schedules movement of the trailers between parking spots and loading docks for loading and unloading of trailer content. However, the embodiments hereof may also be used for intermodal rail and/or port applications and apply to use of a Terminal Operating Systems (TOS), a Warehouse Management System (WMS), and a Transportation Management System (TMS) at any location where trailers are moved autonomously, such as where semi-trailers are used for highway freight movement and where shipping containers are moved on wheeled-chassis as seen as part of port and rail applications, without departing from the scope hereof. For sake of brevity and clarity, the terms YMS and trailers are intended to cover all these options, where the term trailer refers to both semi-trailers used for highway freight movement and to shipping containers that are moved on wheeled chassis that are typically seen as part of port and rail applications.

Mission Control Updated

[0023] Mission control (MC) is a service that manages autonomous vehicles (AVs) in a customer's yard, which may have a staging area for arriving and departing trailers, and a warehouse with loading docks for loading assets to and unloading assets from the trailers. The customer uses a yard management system (YMS) to track and control arrival of trailers and contained assets to the yard, movement of trailers and assets in the yard, and departure of trailers with assets from the yard. Conventionally, to use the AV to move the trailers between parking spots in the staging areas and loading docks at the warehouse, the customer sends certain trailer move requests to MC to be performed by AVs, and schedules other move requests to be performed manually by drivers. MC generates missions for the AVs to perform the received move requests. When the move request is completed successfully, MC indicates to the YMS that the move request was a success (e.g., the move was done or complete). Where the AV is unable to complete the move, MC indicates to the YMS that the move request was rejected (e.g., the move was not done or is incomplete). The rejection occurs for many different reasons, including: a system error, a damaged trailer, criteria falling outside an operating design domain (ODD) of the AVs, inventory issue, issue at the dock such as a dock light not being green, and so on. Move requests that are rejected are rescheduled for manual handling by the YMS, which is often disruptive to already scheduled manual operations when the rejection is delayed, such as when it occurs after unsuccessful attempts to move the trailer by the AV.

[0024] To overcome these problems and advance the art, MC is updated to interact directly with the YMS to provide a more streamlined and transparent AV operation. One key advantage is that the updated MC implements a pull model using an application programming interface (API) of the YMS. MC accesses information within the YMS to determine which move requests may be completed by AVs, and marking other move requests as unsuitable to AVs. MC evaluates each move request prior to selecting for AV operation. Where information within MC indicates that the move request has an issue (e.g., a damaged trailer, criteria falling outside ODD, inventory issue, issue at the dock like a light not being green, etc.) that prevent handling by an AV, MC does not pull the move request from the YMS. At any stage after pulling the move request to attempting and completing the move, MC may return a status to the YMS with a rejection as the result of a system error, a later discovered issue with the move (including those listed previously above) or move completed successfully. MC automatically returns a notice to the YMS indicating a status of the move request. Advantageously, MC automatically selects only move requests that may be handled by an AV based on continuously updated knowledge of the yard and trailers, resulting in fewer move requests being rejected and manually scheduled by the YMS, and less disruption to the manual operation.

[0025] FIG. **1** is a schematic diagram illustrating example remote web based controls **100** for autonomous operation, in embodiments. A customer operates a yard management system **106** (YMS **106**) to manage an inventory **134** that tracks locations of trailers **112** within a yard **104**. Yard **104** has a plurality of parking spots **114** (e.g., staging area parking spots and loading dock parking spots) that may, or may not, have a trailer parked therein. For example, where yard **104** includes a warehouse, certain parking spots **114** are loading docks of the warehouse, and other parking spots **114** are in a staging area from which newly arrived trailers are collected and to which trailers ready for departure are delivered. YMS **106** maintains inventory **134** of trailers **112** within yard **104**, defines which parking spot **114** each trailer is parked in, and generates move requests **108** to schedule movement of trailers **112** within yard **104**, as needed for loading and unloading of assets for example. In certain embodiments, inventory **134** also maintains a trailer database **136** that indicates a status of each trailer **112**, such as trailer type, trailer health (e.g., damage and/or trailer operability), and trailer contents, such as whether trailer **112** is empty or not. In the example of FIG. **1**, trailer **112**(**1**) is parked in parking spot **114**(**1**) and trailer **112**(**2**) is parked in parking spot **114** (**5**).

[0026] An enhanced mission control **120** (MC **120**, which may be an onsite computer server and/or cloud based service) controls operation of at least one AV **102** (AV **102**) in yard **104** to autonomously move trailer **112** from a starting parking spot **114** to a destination spot **114** based on a move request **108** retrieved from YMS **106**. In the example of FIG. **1**, MC **120** controls three AVs **102**(**1**), **102**(**2**), and **102**(**3**) within yard **104**; however, MC **120** may control more or fewer AVs **102** and may concurrently operate AVs within other yards.

[0027] MC **120** includes a puller **122**, an assessor **124**, a scheduler **125**, an updater **127**, a problem resolution tool **132**, and an AV interface **130**, implemented as software that controls at least one processor to implement functionality of MC **120** as described herein. MC **120** also maintains a mission queue **126** for storing missions **128** to be performed by AVs **102**. Puller **122** interacts, directly via an API **107**, with YMS **106** to autonomously detect a new or updated move request **108** within a move queue **110**. Puller **122** invokes assessor **124** to determine whether move request **108** is feasible (e.g., meets certain criteria that indicate that AV **102** is able to perform the move), invokes scheduler **125** to generate a mission **128** stored within mission queue **126**, and updates a corresponding status **129** within move queue **110**. For example, updater **127** automatically updates, via API **107**, status **109** within move queue **110** of YMS **106** to reflect a status of mission **128**. Advantageously, YMS **106** displays the status of both manual moves and autonomous moves such that a user of YMS **106** does not need to switch to viewing a dashboard of MC **120** to see a current status of all moves within yard **104**. Operation of assessor **124** is described in further detail below with reference to FIG. **2**. Problem resolution tool **132** is invoked to resolve moves that cannot be completed, such as when a pick-up spot is empty or a destination spot is already occupied.

[0028] In one example of operation, puller **122** detects move request **108**(**1**) within mission queue **126** of YMS **106**, where move request **108**(**1**) requests movement of trailer **112**(**1**) from parking spot **114**(**1**) to parking spot **114**(**4**). Puller **122** invokes assessor **124** to determine the move is feasible, whereby assessor **124** determines (a) that trailer **112**(**1**) is parked in parking spot **114**(**1**), (b) that parking spot **114**(**4**) is empty, (c) that AV **102**(**1**) is available and capable of moving trailer **112**(**1**) from parking spot **114**(**1**) to parking spot **114**(**4**), and (e) that the move meets certain criteria (e.g., the trailer in undamaged, criteria does not fall outside ODD, there is no inventory issue, there are no issue at the dock such as the dock light not being green, etc.). For example, AV **102** may be configured to move a forty-eight-foot dry van, but is not configured to move a container trailer. Puller **122** then invokes scheduler **125** to generate mission **128**(**2**) with directives that control AV **102**(**1**) to move trailer **112**(**1**) from parking spot **114**(**1**) to parking spot **114**(**4**), stores mission **128**(**2**) in mission queue **126**, sets a corresponding status **129**(**2**) to "in progress", and invokes updater **127** to update corresponding status **109**(**1**) of YMS **106**.

[0029] In another example of operation, where move request **108**(**2**) requests movement of trailer **112**(**1**) from parking spot **114**(**1**) to parking spot **114**(**4**), puller **122** detects move request **108**(**2**) within mission queue **126** of YMS **106**, and invokes assessor **124** to determine whether the move is feasible for AV **102**. In this example, assessor **124** determines that parking spot **114**(**4**) is not empty based on evidence recently captured by a drive-by of parking spot **114**(**4**) by one AV **102**, which indicated that another trailer is parked in parking spot **114**(**4**). Accordingly, assessor **124** invokes updater **127** to interact with YMS **106** and set corresponding status **109**(**2**) to indicate that the move is not possible. In certain embodiments, updater **127** may provide additional reasons, such as indicating that the move is not possible because parking spot **114**(**4**) is not empty, and may further indicate the identity of the trailer currently in parking spot **114**(**4**). Assessor **124** may also send a notification to YMS **106** (e.g., to a user of YMS **106**) indicating that the trailer move is not possible and that an alternative destination spot is needed. In certain embodiments, assessor **124** invokes problem resolution tool **132** to determine whether another nearby parking spot (e.g., parking spot **114**(**6**)) would be acceptable for parking trailer **112**(**1**), whereby problem resolution tool **132** updates mission **128**(**2**) to indicate parking spot **114**(**6**) as the destination spot for trailer **112**(**1**).

[0030] AV **102** collects information of parking spots **114** as it maneuvers within yard **104**. In certain embodiments, AV **102** includes functionality to automatically search nearby for parking spots **114** that are empty, and/or to identify trailers in nearby parking spots to allow MC **120** to updates its mission **128** accordingly. For example, where a destination spot is occupied, AV **102** identifies an empty nearby spot, and requests MC **120** to update its mission **128** to deposit trailer **112** in parking spot **114** that was found to be empty. MC **120** may also update inventory **134** of YMS **106** based on a status of parking spots **114** detected by AVs **102** during normal operation within yard **104**. For example, as AV **102** maneuvers within yard **104**, AV **102** captures information (e.g., using one or more of its cameras, its LIDAR, and its radar, etc.) indicative of the state of yard **104**. For example, AV **102** may capture information such as trailer IDs **218** and a status of any trailer detected in parking spots **114** passed by AV **102**. The trailer information may indicate whether trailer **112** has damaged would prevent autonomous movement of the trailer, or whether the positioning of trailer **112** within parking spot **114** prevents autonomous movement of the trailer, and so on. AV **102** may also collect other yard information such as unidentified or unexpected objects (e.g., dropped pallets, trash, tools, etc.) that would impede movement of AVs **102** and/or trailers **112** within yard **104**, etc. Advantageously, inventory **134** is automatically updated during normal operation of AVs **102** within yard **104** and is therefore kept updated without requiring additional control of the AVs **102**.

[0031] In one operational example, as AV **102** reaches a destination spot of mission **128**, when AV **102** determines that the destination spot is not empty, AV **102** notifies MC **120** to request an empty nearby spot. MC **120** invokes problem resolution tool **132** to identify an acceptable nearby parking spot and update move request **108** with the destination spot. Problem resolution tool **132** then generates a updated mission **128** to deposit trailer **112** in the nearby parking spot, and sends updated mission **128** to AV **102**. When problem resolution tool **132** cannot find an empty nearby spot, problem resolution tool **132** instructs AV **102** to search nearby to find an empty spot. For example, AV **102** processes captured imagery using deep learning algorithms to identify an empty nearby spot. When AV **102** finds an empty nearby spot, AV **102** sends the empty nearby spot to problem resolution tool **132**, which updates move request **108** and mission **128** to deposit trailer **112** in the empty nearby spot, and sends updated mission **128** to AV **102**, causing AV **102** to deposit trailer **112** in the empty nearby spot. Problem resolution tool **132** also notifies YMS **106** that the trailer is parked in the nearby spot and not the intended destination. Problem resolution tool **132** may also update YMS **106** to identify the trailer, based on the captured information from AV **102**, within the originally intended destination spot.

[0032] When AV **102** determines that a pick-up spot of a current mission **128** (corresponding to move request **108**) is

empty, indicating the trailer is not found, AV **102** notifies MC **120** that the pick-up spot is empty. Accordingly, MC **120** invokes problem resolution tool **132** to search YMS **106** for an updated location of the trailer identified in the corresponding move request **108**, and when found, problem resolution tool **132** updates the pick-up location in the corresponding move request **108**, generates a new mission **128** based on the updated pick-up spot, and sends it to AV **102**, thereby causing AV **102** to traverse to the updated pick up location to collect the identified trailer. When problem resolution tool **132** does not find a new location for the requested trailer, problem resolution tool **132** instructs AV **102** to search nearby spots for the requested trailer. For example, AV **102** processes captured imagery using deep learning algorithms to identify nearby trailers, and when AV **102** finds the trailer (e.g., based on a trailer identifier) in a nearby spot, AV **102** sends the nearby spot to problem resolution tool **132**. Accordingly, problem resolution tool **132** generates a new mission **128** based on the nearby spot, sends the new mission **128** to AV **102**, which causes AV **102** to collect the trailer from the nearby spot.

[0033] AV interface **130** processes mission queue **126** and sends missions **128** to the assigned AVs **102**. For example, AV interface **130** sends mission **128**(**2**) to the AV **102**(**1**) to instruct AV **102**(**1**) to move trailer **112**(**1**) from parking spot **114**(**1**) to parking spot **114**(**4**). If, during mission **128**(**2**), AV **102**(**1**) detects an anomaly preventing the trailer move, AV **102**(**1**) reports the anomaly to AV interface **130**, which may attempt to resolve the anomaly (e.g., invoking a remote support platform—remote assist) or may invoke updater **127** to update the corresponding status **129**(**2**) and status **109**(**3**) of YMS **106**.

[0034] Advantageously, puller **122** and assessor **124** cooperate to select only ones of move requests **108** that may be performed by AVs **102**. Where assessor **124** determines that move request **108** cannot be performed autonomously, assessor **124** sets the corresponding status **109** to indicate that autonomous move is not possible. Advantageously, YMS **106** is updated to reflect the status of each move request **108** such that these moves may be handled manually.

[0035] A further advantage of using API **107** to interact with YMS **106** directly, is that assessor **124** may improve operational efficiency of AVs **102** by selecting and scheduling ones of move requests **108** that minimize wasted movement of AVs **102**. For example, missions **128** may be ordered within mission queue **126** to reduce wait times and minimize movement of each AV **102** between trailer moves. For example, within the defined movement windows of move requests **108**, assessor **124** may select and schedule missions **128** based on the locations of starting and ending parking spots **114** to minimize movement of each AV **102** when not moving trailers.

[0036] In another example of operation, move request **108** indicates that an identified trailer **112** is to be moved to a destination spot, but does not specify the pick-up spot. Since the pick-up spot is undefined, MC **120** invokes problem resolution tool **132** to search YMS **106** (e.g., a yard inventory stored at YMS **106**) for the identified trailer to determine a pick-up spot. When the pickup spot is found, problem resolution tool **132** generates a new mission **128** to cause AV **102** to move the identified trailer from the pick-up spot to the destination spot.

[0037] In another example of operation, move request **108** indicates that a trailer type is to be moved to a destination spot, but does not identify the trailer or specify the pick-up spot. For example, move request **108** may state "move an empty **28'** dry van to a destination spot." Accordingly, MC **120** invokes problem resolution tool **132** to search for the trailer type in trailer database **136** and then to determine the location of matching trailers **112** within inventory **134**. When a matching trailer is found, problem resolution tool **132** assigns the found trailer type to move request **108** and generates mission **128** to cause AV **102** to move the matching trailer from its pick-up spot (e.g., the current location of the matching trailer) to the destination location. The AV uses imagery and deep learning algorithms to identify the trailer at the pick-up spot to verify that the matching trailer is parked in the pick-up spot. When the identified trailer does not correspond to the matched trailer, AV **102** sends the trailer identifier to problem resolution tool **132** and requests a new location for another trailer matching the trailer type from problem resolution tool **132**. When problem resolution tool **132** finds another trailer matching the requested trailer type in the inventory, problem resolution tool **132** updates mission **128** with a new trailer identifier and a new pick-up spot. Updated mission **128** is sent to AV **102** and causes AV **102** to proceed to the new pick up spot to collect the trailer. When problem resolution tool **132** does not find a match within trailer database **136**, problem resolution tool **132** instructs AV **102** to search nearby and identify parked trailers **112**. AV **102** sends the identify of found trailers to problem resolution tool **132**, which determines whether the found trailer matches the required trailer type of move request **108**. When the identified trailer matches the required trailer type, problem resolution tool **132** generates a new mission **128** based on a new pick-up spot (e.g., received from AV **102** with the trailer ID) of the matching trailer, sends the new mission **128** to AV **102** to cause AV **102** to move the matching trailer from the new pick-up spot to the destination spot.

Weather Notifications

[0038] MC **120** also includes a weather API **140**, a weather monitor **144** and an operational schema **146**. Weather API **140** that allows MC **120** to ingest weather data **142** (e.g., weather forecasts, local weather reports, etc.). Operational schema **146** defines an operational status of MC **120** and may include an operational status of AVs **102**, and a predicted operational status of AVs **102**. Operational schema **146** is used by other components of MC **120** and defines operation parameters of MC **120** and AVs **102**.

[0039] Weather monitor **144** processes weather data **142** to determine a current and/or future operational status of AVs **102** and updates operational schema **146** accordingly. Weather data **142** may include one or more of a weather forecast received from a third party weather service for an area including yard **104**, a weather report (e.g., current weather conditions) from a local weather station, and a status report defined from one or more sensors of AV **102** and included in telemetry data received via AV interface **130**.

[0040] Weather monitor **144** processes weather data **142** and updates operational schema **146** of MC **120** to define current and future operability of AVs **102**. For example, where weather data **142** indicates heavy snow between 3 PM and 6 PM, weather monitor **144** updates operational schema **146** to indicate that AVs **102** are not operable between 3 PM and 6 PM. Similarly, where weather monitor **144** determines from weather data **142** that heavy rain has started falling,

weather monitor **144** updates operational schema **146** to indicate operation of AVs **102** is not possible. In another example, where weather monitor **144** determines from weather data **142** that mist is reducing visibility, weather monitor **144** updates operational schema **146** to indicate operation of AVs **102** is restricted and/or not possible.

[0041] Where weather monitor **144** changes operational schema **146**, such as when weather data **142** first indicates heavy snow between 3 PM and 6 PM, weather monitor **144** may also send a weather notification to YMS **106** (e.g., users thereof) and/or operators of MC **120**. Advantageously, predicted outage of autonomous trailer movement allows the user of YMS **106** to plan alternative arrangements, such as to provide drivers to ride in AVs **102** during reduced visibility, and/or to use manual (non-autonomous) yard tractors for trailer movement during weather conditions in which AVs **102** cannot operate.

Move Capability

[0042] FIG. **2** is a block diagram illustrating operation of assessor **124** of FIG. **1** in further example detail. Assessor **124** may indicate that move request **108** is not feasible for a variety of reasons, and accordingly the move request is rejected. Advantageously, assessor **124** determines that move request **108** is not feasible at the earliest opportunity by evaluating many known facts that could influence the corresponding mission **128**. The sooner move request **108** is rejected, the sooner it may be reassigned and completed, resulting in better efficiency within yard **104**. Additionally, data known before executing mission **128** may result in faster completion of the move request **108**. Accordingly, MC **120** collects data continuously and throughout every area in yard **104** as AVs **102** operate, and MC **120** uses the collected data to inform assessor **124** and the planner of missions **128**.

[0043] Each AV **102** collects data that is sent to a data collector **202** of MC **120**. For example, data collector **202** may be part of a cloud service that aggregates this data and makes it available to other components of remote web based controls **100** to inform behavior. Data collector **202** receives updates from YMS **106** that include trailer information **210**, inventory information **220**, and facility information **230**. Data collector **202** may also receive trailer information **210**, inventory information **220**, and facility information **230** from AVs **102** operating within yard **104**. In certain embodiments, MC **120** stores collected data in an MC database **250** and thereby automatically detects changes indicated by trailer information **210**, inventory information **220**, and facility information **230**. Data collector **202** may communicate detected changes to other components of remote web based controls **100** and to YMS **106**. Data collector **202** may also pull information from MC database **250** and/or YMS **106**. The following description provides examples of how certain data points are used.

[0044] Trailer information **210** relates to information about trailer **112** that is collected by any AV **102** that is near trailer **112**. Trailer information **210** may be used by assessor **124** to determine when move request **108** are not feasible for a particular trailer prior to start of a corresponding mission **128**. Trailer information **210** may include gladhand data **212**, trailer body and positioning data **214**, trailer tandem data **216**, and a trailer ID **218**. Trailer information **210** may also indicate where trailer **112** is damaged. For example, trailer information **210** may indicate one or more of roof damage, sidewall damage, connector damage, and other issues that prevent autonomous handling of trailer **112** and/or indicate that trailer **112** is in need of repair. In certain embodiments, trailer information **210** is sent to update YMS **106** and corresponding requested autonomous operations on trailer **112** that cannot be completed are returned to YMS **106** (e.g., rejected by MC **120**). Gladhand data **212** may include type, position, and state of gladhands (e.g., from images captured by AVs **102**) of gladhands on any trailer in yard **104**. Accordingly, assessor **124** may reject moves of trailer **112** when it has a damaged gladhand without requiring an AV to drive to the trailer, since the damage was reported by another AV **102** when passing trailer **112**. Further, gladhand data **212** may help with pose estimation for autonomous connections during execution of the corresponding mission **128**. Trailer body and positioning data **214** indicates a status of the trailer body and its position with a parking spot **114** for example. Trailer body and positioning data **214** may be collected by AVs passing the trailer and/or reported by YMS **106**. Assessor **124** may use trailer body and positioning data **214** to determine when trailer **112** is damaged or in a position that prevents hitching by AV **102**, whereby assessor **124** rejects the move request **108** prior to generating the corresponding mission **128** or requiring an AV to visit the trailer. Trailer tandem data **216** defines the position of tandems on trailer **112** and may be used by motion planning of MC **120** when preparing mission **128** and thus before executing the mission such that time of the in-move calculation is reduced. In certain embodiments, AV **102** processes captured images to determine trailer ID **218**. In other embodiments, MC **120** processes images received from AV **102** to determine trailer ID **218**.

[0045] Inventory information **220** defines inventory parameters that allow assessor **124** to assess move request **108** before generation and/or execution of the corresponding mission **128**. For example, assessor **124** rejects move requests **108** or initiates exception planning when inventory information **220** indicates that trailer **112** is not parked in the expected parking spot **114** and/or when the intended destination spot **114** is not empty. Inventory information **220** may include origin location **222** and destination location **224** for identified trailer **112**. For example, as AVs **102** drive around yard **104**, they may detect and report trailer identities and locations, including trailer body and positioning data **214**, whereby data collector **202** updates corresponding inventory information **220** in MC database **250** and/or YMS **106**. Origin location **222** allows assessor **124** to confirm that the expected trailer **112** is in the origin location or plan a contingency. Destination location **224** allows assessor **124** to confirm the destination location is unoccupied or plan a contingency.

[0046] Facility Information **230** includes information about conditions of yard **104** and may be received from YMS **106** and/or received from AVs **102** as they traverse yard **104** and/or from remote operators when responding to assist requests. Facility information **230** allows assessor **124** to determine when AV **102** is unable to perform the move prior to generation or execution of the corresponding mission **128**, resulting in earlier rejection of move request **108**, contingency planning, or exception communication. Facility information **230** may include route planning **232**, apron conditions **234**, and dock light status **236**. Route planning **232** allows MC **120** to adjust missions **128** within specified parameters to optimize routes ahead of time. For example, when mission **128** tasks AV **102** to autonomously select an

available parking spot, AV **102** may select one that is slightly further away to minimize global execution time of mission **128** and a subsequent mission. MC **120** evaluates (e.g., continually, at intervals, and/or in response to events) one or more of mission queue **126**, trailer information **210**, inventory information **220**, and facility information **230** to reorganize (e.g., optimize efficiency) missions **128** of mission queue **126** based on changes, collected by data collector **202**, to the state of yard **104**. For example, MC **120** instructs AV **102** to modify its current course (e.g., mid-mission) to improve efficiency of tasks being performed within yard **104**. Apron conditions **234** allows MC **120** to plan routes around issues developed in the apron. Dock light status **236** allows assessor **124** to determine when a red or damaged dock light would prevent completion of the corresponding mission **128**, communicate and exception, and/or move the mission **128** down mission queue **126** once, such that a subsequence mission **128** may be started. When the prevented mission is attempted again, when the same condition (e.g., red or damaged dock light) remains, the mission is moved down mission queue **126** again and the subsequent mission **128** executed.

Remote Support Platform (RSP)

[0047] FIG. **3** is a block diagram illustrating MC **120** of FIG. **1** with a remote support platform **302** that allows a remote operator **320** to use an operator console **322** to remotely monitor and control any one AV **102**, in embodiments. In certain embodiments, operator console **322** runs a browser that interacts with a RSP user interface **303** of remote support platform **302** and/or a user interface of MC **120**. Operator console **322** may communicate with remote support platform **302** to send a telemetry request **326** to AV **102** to monitor operation thereof. Remote support platform **302** may also request operator console **322** assist AV **102**, such as when AV **102** requests assistance during autonomous operation, whereby remote support platform **302** sends telemetry request **326** to AV **102** to initiate telemetry feed **324**. For example, where AV **102** is unable to couple (hitch) with a trailer, AV **102** may requests assistance after three unsuccessful autonomous attempts, whereby remote operator **320** uses operator console **322** to control RSP user interface **303** to send telemetry request **326** to AV **102**, receive telemetry feed **324** (e.g., including at least part of telemetry data **304**) from AV **102**, and may send one or more control directives **328** to AV **102** via further interaction with RSP user interface **303** via operator console **322**. In another example, when AV **102** is unable to resolve a detected object within an image with sufficient confidence, AV **102** requests assistance, whereby remote operator **320** uses operator console **322** to receive telemetry data **304** in telemetry feed **324** from AV **102** and interact with RSP user interface **303** to resolve the object. Further, operator console **322** may display AV status and events, including weather, at a yard as they occur, allowing operator **320** to make operational decisions and assist a customer (e.g., owner of yard **104**) as needed. Advantageously, the AVs may be remotely monitored and assisted from anywhere in the world with a stable internet connection via remote support platform **302**.

[0048] Remote support platform **302** is a full-stack application (e.g., a full end-to-end software development) that allows operators to monitor and assist AVs **102** performing missions **128** and interact with MC **120** to generate missions **128** that command the AVs as needed. A conventional remote monitoring and control tool provides individual control of an AV, but does not scale to allow control of an entire fleet of AVs and does not have the ease-of-use and reach as a web application to provide remote monitoring and control of any AV via a stable internet connection. Particularly, remote support platform **302** allows any remote operator **320** to provide assistance to any AV **102**.

[0049] FIG. **4** is a block diagram illustrating example detail of AV **102** of FIGS. **1** and **3**, in embodiments. FIGS. **3** and **4** are best viewed together with the following description.

[0050] AV **102** includes a battery **402** for powering components of AV **102** and a controller **406** with at least one digital processor **408** communicatively coupled with memory **410** that may include one or both of volatile memory (e.g., RAM, SRAM, etc.) and non-volatile memory (e.g., PROM, FLASH, Magnetic, Optical, etc.). Memory **410** stores a plurality of software modules including machine-readable instructions that, when executed by the at least one digital processor **408**, cause the at least one digital processor **408** to implement functionality of AV **102** as described herein to operate autonomously within autonomous yard **104** under direction from MC **120**.

[0051] When AV **102** is an electric tractor, AV **102** also includes at least one drive motor **412** controlled by a drive circuit **414** to mechanically drive a plurality of wheels (not shown) to maneuver AV **102**. AV **102** also includes a location unit **416** (e.g., a GPS receiver) that determines an absolute location and orientation of AV **102**, a plurality of cameras **418** for capturing images of objects around AV **102**, and at least one Light Detection and Ranging (LIDAR) device **420** (hereinafter LIDAR **420**) for determining a point cloud about AV **102**. Location unit **416**, the plurality of cameras **418**, and the at least one LIDAR **420** cooperate with controller **406** to enable autonomous maneuverability and safety of AV **102**. AV **102** includes a fifth wheel (FW) **422** for coupling with trailer **112** and a FW actuator **424** controlled by controller **406** to position FW **422** at a desired height. In certain embodiments, FW actuator **424** includes an electric motor coupled with a hydraulic pump that drives a hydraulic piston that moves FW **422**. However, FW actuator **424** may include other devices for positioning FW **422** without departing from the scope hereof. AV **102** may also include an air actuator **438** that controls air supplied to trailer **112** and a brake actuator **439** that controls brakes of AV **102** and trailer **112** when connected thereto via air actuator **438**.

[0052] Controller **406** also determines a trailer angle between AV **102** and trailer **112** based on one or both of a trailer angle measured by an optical encoder **404** positioned near FW **422** and mechanically coupled with trailer **112** and a point cloud captured by the at least one LIDAR **420**.

[0053] Controller **406** may implement a function state machine that controls operation of AV **102** based upon commands (requests) received from MC **120**. In the example of FIG. **3**, MC **120** receives a request (e.g., via an API, and/or via a GUI used by a dispatch operator) to move trailer **112**(**1**) from parking spot **114**(**1**) to plurality of parking spots **114**(**4**). Once this request is validated, MC **120** invokes a mission planner (e.g., a software package, not shown) that computes mission **128** for AV **102**(**2**). For example, the mission plan is an ordered sequence of high level primitives to be followed by AV **102**, in order to move trailer **112**(**1**) from parking spot **114**(**1**) to parking spot **114**(**4**). The mis-

sion plan may include primitives such as drive along a first route, couple with trailer **112**(**1**) in parking spot **114**(**1**), drive along a second route, back trailer **112**(**1**) into parking spot **114**(**4**), and decouple from trailer **112**(**1**).

[0054] Function state machine includes a plurality of states, each associated with at least one software routine (e.g., machine-readable instructions) that is executed by the at least one digital processor **408** to implements a particular function of AV **102**. Function state machine may transitions through one or more states when following the primitives from MC **120** to complete the mission plan.

[0055] Controller **406** may also include an articulated maneuvering module, implemented as machine-readable instructions that, when executed by the at least one digital processor **408**, cause the at least one digital processor **408** to controls drive circuit **414** and steering actuator **425** to maneuver AV **102** based on directives from MC **120**.

[0056] Controller **406** may also include a navigation module that uses location unit **416** to determine a current location and orientation of AV **102**. Navigation module may also use other sensors (e.g., camera **418** and/or LIDAR **420**) to determine the current location and orientation of AV **102** using dead-reckoning techniques.

[0057] In one operational example of FIG. **3**, AV **102**(**2**) encounters a problem when performing mission **128**(**2**) to move trailer **112**(**1**) from parking spot **114**(**1**) to parking spot **114**(**4**). During a drive by of parking spot **114**(**4**) to verify that the parking spot is clear, due to light rain and poor lighting, AV **102**(**2**) cannot achieve sufficient confidence that parking spot **114**(**4**) is empty prior to reversing trailer **112**(**1**) into parking spot **114**(**4**). Accordingly, AV **102**(**2**) stops and requests assistance of remote support platform **302**.

[0058] Remote support platform **302** determines that operator **320**(**1**) is available and sends a request for remote assistance to operator console **322**(**1**). Remote support platform **302** determines a bandwidth of the communication between operator console **322**(**1**) and MC **120**, and optionally between MC **120** and AV **102**(**2**), and instructs AV **102**(**2**) to send telemetry data **304**(**2**) to operator console **322**(**1**) via remote support platform **302**. Since AV **102**(**2**) has multiple cameras, and other sensors, the volume of telemetry data **304** generated by AV **102**(**2**) (and any other AV) is large.

Video Bakery

[0059] On AV **102**, cameras **418** output encoded video streams that are stored in video buffers **419** within memory **410** for example. However, video buffers **419** and the video streams stored therein are difficult to decode and manipulate to determine a still image, which may for example be needed to resolve an issue requiring assistance. Particularly, the video feeds require complex decoding, to retrieve a still image for example. Accordingly, AV **102** is equipped with a video bakery **330** that provides a bakery interface **452** that allows other components (e.g., a telemetry application **440** and/or a remote assistance application **442**) of AV **102** to retrieve still images and video streams with a variety of configuration parameters (e.g. setting frame rate, output image/video resolution, compression/encoding type). For example, remote assistance application **442** may generate a telemetry request **326** that instructs video bakery **330** to generate output stream **460** for use in telemetry data **304** being sent to operator console **322**(**1**). Accordingly, video bakery **330** may crop and scale frames of still images and video streams to generate output stream **460**.

[0060] Bakery interface **452** receives telemetry request **326** identifying an area of interest, and defining a maximum bandwidth for telemetry. For example, telemetry application **440** instructs bakery interface **452** to generate and send a telemetry feed **324** to operator console **322**(**1**) based on a request for assistance generated by remote assistance application **442**. Bakery interface **452** may invokes a feed select algorithm **458** to determine ones of the plurality of cameras **418** positioned to provide relevant information for telemetry feed **324**. In one example, where remote assistance application **442** indicates assistance is required to drive AV **102** forwards, feed select algorithm **458** selects forward facing ones of the plurality of cameras **418**. In another example, where remote assistance application **442** indicates assistance is required to evaluate whether parking spot **114**(**4**) is empty, feed select algorithm **458** selects ones of the plurality of cameras **418** facing parking spot **114**(**4**) based on the current location and orientation of AV **102** and further requests a high resolution still image of parking spot **114**(**4**).

[0061] Bakery interface **452** determines a frame rate and/or resolution, based on the maximum bandwidth defined within telemetry request **326**, for output stream **460** and/or still image **462** that are included in telemetry feed **324**. Bakery interface **452** then invokes stream manipulator **456** to generate output stream **460** from one or more selected cameras **418** and invokes a still image generator **454** to process one or more image feeds from the selected cameras **418** to generate a still image **462** of the area of interest (e.g., the parking spot **114**(**4**)). In one example of operation, telemetry request **326** identifies one or more cameras **418** of AV **102**, whereby video bakery **330** generates output stream **460** and/or still image **462** based on the identified cameras and maximum bandwidth. In another example of operation, telemetry request **326** defines a point or region in three dimensional space around AV **102**, whereby video bakery **330** uses feed select algorithm **458** to automatically select one or more cameras **418** based on the point or region. Feed select algorithm **458** uses camera calibration values (e.g., extrinsic and intrinsic calibration values) to project the input point/region in space into the camera frame to determine if the point/region is within a field of view of each camera on the vehicle. Bakery interface **452** then invokes a stream manipulator **456** to process image feeds from the manually or autonomously selected cameras **418** and generate an output stream **460** and/or still image **462** at the defined maximum resolution and the maximum frame rate. Accordingly, stream manipulator **456** may convert the frame rate of the image feeds from the selected cameras **418** into a frame rate less or equal to the maximum frame rate. In one example of operation, stream manipulator **456** generates output stream **460** as a composite grid of feeds, where each grid element includes at least part of a field of view of a different one of the selected cameras **418**. For example, remote support platform **302** may select four cameras **418** that provide a **360** degree view around AV **102**, whereby output stream **460** is divided into a two-by-two grid, and each grid element includes at least part of a field-of-view of a different one of the four selected cameras **418**. For example, one or more of the grid element includes a full field of view from the corresponding camera **418**. In another example of operation, to maximize usefulness of the telemetry feed **324**, one or more of the grid element includes a partial field of view

from the corresponding camera **418**, where the partial field of view is selected based on the point/region defined within telemetry request **326**. For example, where a forward-right facing camera provides a three-hundred-and-forty-degree field-of-view, stream manipulator **456** may select a ninety-degree sub-field-of-view that includes the defined point/region that corresponds to a trailer identifier that requires assistance to resolve. Accordingly, less relevant portions of the image feed from the camera are automatically excluded. Advantageously, output stream **460** provides the most relevant information without exceeding the maximum bandwidth.

[0062] Video bakery **330** sends output stream **460** and/or still image **462** to one or both of telemetry application **440** and remote assistance application **442**, such that it is included in telemetry feed **324**, which is sent to operator console **322**(**1**). Video bakery **330** operates only as needed to generate output stream **460** and/or still image **462**. Advantageously, video bakery **330** performs complex video decoding to get image frames from video buffers **419** and thereby facilitates use of images and video streams by other components of AV **102** and/or MC **120** without requiring knowledge of the encoding/encryption algorithms used within AV **102**.

Vehicle Credential Management (VCM)

[0063] FIG. **5** shows one example VCM system **500** that facilitates autonomous configuration of AV **102**, in embodiments. VCM system **500** includes a VCM server **530**, a configuration database **540**, and a credential vault **550**. Configuration database **540** is for example implemented by a PARSE server. Credential vault **550** is for example implemented by a HashiCorp Vault. In certain embodiments, configuration database **540** and credential vault **550** are combined into a single database that may be part of VCM server **530** and/or external to VCM server **530**.

[0064] AV **102** (e.g., a new AV, or new deployment of a fleet of AVs) is delivered to yard **104** with a standard, generic, software configuration, and includes a unique AV identifier **502**, AV certificate **504**, and an AV configuration manager **508**. AV **102** also includes communication functionality that allows AV **102** to connect to a local network (e.g., a wireless LAN such as Wi-Fi). Unique AV identifier **502** is for example a hardware Universally Unique Identifier (UUID) that is unique to each AV **102** and AV certificate **504** provides authenticity to AV **102**. To enable functionality of AV **102** for operation at yard **104**, AV **102** requires updating with a specific software configuration (e.g., a software stack **510** that includes specific software, backend servers communication information, and/or credentials). AV configuration manager **508** is invoked during boot of controller **406** to communicate with VCM server **530** and automatically update software stack **510** if needed.

[0065] Conventionally, AV **102** configuration was performed on each AV individually based on its intended use. For example, each AV would be manually configured with a specific software load to provide capability for a specific customer at a specific yard. In another example, an operator would wirelessly connect to one AV and manually control loading of specific software and/or credentials into memory of the AV.

[0066] VCM server **530** includes a user interface **532**, and configuration manager **534**, and an AV API **536**. AV API **536** is for example a custom GraphQL API. VCM system **500** allows cloud-based AV configuration, whereby each AV **102** updates itself autonomously under control of VCM server **530** to provide a more streamlined and scalable AV configuration solution from a centralized location. For example, each AV in a fleet of new AVs deployed to a yard is easily provisioned and/or updated with a required software stack **510** under control of VCM server **530**, where configuration database **540** includes a configuration definition **542** defining software stack **510** needed to provide the intended function of AV **102**.

[0067] An operator interacts with user interface **532** of VCM server **530** to create or change configuration definition **542** in configuration database **540** for an unique AV identifier **502**. Configuration definition **542** is associated with at least one of a plurality of software definitions **552**, corresponding credentials **554**, and corresponding software code **556** that are stored within credential vault **550**. In certain embodiments, software code **556** is stored elsewhere and manually downloaded into each AV **102** as needed.

[0068] Configuration definition **542** defines ones of software definition **552**, credentials **554**, and corresponding software code **556** that are downloaded to AV **102** to form software stack **510**. Software definition **552** identifies software code **556** (e.g., software including machine readable instructions) and/or configuration data (e.g., operational parameters and definitions for one or more software modules) that may be loaded into software stack **510** of AV **102** to provide a specific functionality of AV **102**. Accordingly, an operator updates one or more of configuration definition **542**, credentials **554**, and/or software code **556** to update functionality of one or more AVs **102**. Advantageously, at next initialization of each AV **102** (e.g., at next boot), AV configuration manager **508** communicates with VCM server **530** via AV API **536** to autonomously receive software definition **552**, corresponding credentials **554**, and/or software code **556** associated with configuration definition **542**, where configuration definition **542** matches unique AV identifier **502** of AV **102**. In certain embodiments, AV configuration manager **508** may also be triggered to update software stack **510** by receiving a command sent from VCM system **500**.

[0069] Configuration manager **534** may implement a dashboard, displayed by user interface **532**, that allows the operator to interactively create and/or update configuration definition **542** as needed. In certain embodiments, configuration manager **534** includes intelligence that facilitates the creation and/or update of configuration definition **542** based on software definitions **552** and credentials **554** stored within credential vault **550**, thereby ensuring software stack **510** is configured correctly and up to date. Software developers may add software definition **552** to credential vault **550**, and credentials **554** may be updated as needed (e.g., new credentials generated as existing credentials expire). Accordingly, to deploy new software code **556**, an operator updates configuration definition **542**, whereby AV **102** then autonomously updates its software stack **510** by interacting with VCM server **530** via AV API **536**.

[0070] In certain embodiments, user interface **532** and configuration manager **534** cooperate to implement a full featured UI that allows an operator to manage fleets of AVs. For example, user interface **532** may display an overview of AVs **102** that are running each version of software and where they are deployed, where AVs **102** may be assigned to groups for easier management, such as allowing the operator

to make configuration changes to all AVs in the group based on certain criteria (e.g., hardware, site, etc.). User interface **532** may allow the operator to rotate keys and certificates across one or more AVs, and allow the operator to compare software versions and configurations on different AVs. User interface **532** may also allow the operator to update credentials, configuration, and corresponding/compatible software collectively. For example, where a software program/package is updated (e.g., to a new version), user interface **532** allows the operator to update relevant AVs (e.g., AVs that include the old version of the software) with updated configuration definitions **542** that cause each AV **102** to automatically interact with VCM server **530** to retrieve the associated software definition **552**, credentials **554**, and updated software code **556**.

[0071] Credential vault **550** securely stores many software definitions **552** and many credentials **554**, and each configuration definition **542** within configuration database **540** defines which of these software definition **552** and credentials **554** are to be loaded in the corresponding AV **102**. Configuration definition **542** may also define which backend servers the AV should communicate with, and so forth.

[0072] In certain embodiments, configuration definition **542** defines basic version information associated with each AV. For example, configuration definition **542** defines the required capabilities of AV **102** to function within yard **104**. In certain embodiments, configuration manager **534** processes configuration definition **542** to determine needed software code **556** and credentials required by AV **102**, and then updates configuration definition **542** to reference one or more credentials **554** and/or software definition **552** stored in credential vault **550**. Accordingly, configuration definition **542** defines needed software definition **552** and credentials **554** of AV **102**.

[0073] In certain embodiments, AV configuration manager **508** is invoked at each boot of controller **406** and sends a registration request **560** to configuration manager **534** via AV API **536**. Registration request **560** includes unique AV identifier **502**, AV certificate **504**, and a current location **506** of AV **102**. For example, AV configuration manager **508** determines current location **506** of AV **102** using location unit **416**. Configuration manager **534** authenticates registration request **560**, and thus AV **102**, based on registration request **560**. For example, configuration database **540** may defines a list of valid unique AV identifiers **502** in association with a geographic area of yard **104**. For AV **102** to be authenticated, configuration manager **534** verifies that unique AV identifier **502** is listed within configuration database **540**, and verifies one or more of current location **506** is within the defines geographic areas of yard **104**, and that AV certificate **504** is authentic and current. In certain embodiments, configuration manager **534** verifies that registration request **560** was received via an allowed network (e.g., the wireless LAN at yard **104**).

[0074] In one example of operation, at startup of AV **102**, AV configuration manager **508** sends registration request **560** including at least unique AV identifier **502** to configuration manager **534** via AV API **536**. Once registration request **560** and AV **102** are authenticated, configuration manager **534** retrieves configuration definition **542** from configuration database **540** based on unique AV identifier **502**, and sends configuration definition **542** to AV configuration manager **508** in a configuration status message **562**.

[0075] AV configuration manager **508** compares configuration definition **542** to software stack **510** to determine whether any updates to software stack **510** are needed. Where software updates are needed, AV configuration manager **508** sends an update request **564** to configuration manager **534** via AV API **536**, where update request **564** identifies ones of software definition **552** and/or credentials **554** required by AV **102**. In response, configuration manager **534** sends requested ones of software definitions **552** and/or credentials **554** to AV configuration manager **508** via AV API **536** as a configuration message **566**. AV configuration manager **508** updates software stack **510** with the received software definitions **552** and/or credentials **554**. Accordingly, software stack **510** is autonomously updated by AV **102** with needed software definition **552** and credentials **554** for operation at yard **104**.

[0076] Since credentials may be loaded during booting of controller **406**, it is unnecessary to embed the credentials within software modules. In certain embodiments, credentials **554** are stored in volatile memory of controller **406** and are lost when controller **406** powers down. In another embodiment, AV configuration manager **508** deletes credentials **554** from software stack **510** when AV **102** shuts down. Advantageously, where credentials **554** are not stored in non-volatile memory (or are specifically deleted), the security risk of credentials being copied is reduced. A further advantage of this solution is that credentials **554** are easily updated since credentials **554** are loaded from VCM server **530** at each boot, thereby allowing credentials to be rotated as needed.

[0077] In certain embodiments, as part of increased security, configuration manager **534** only allows registration to occur when last registered date (e.g., stored in configuration database **540** in association with unique AV identifier **502**) is null or indicates that a previous registration was within a predefined amount of time (e.g., 1 week). In these embodiments, when AV **102** is authenticated, configuration manager **534** stores a timestamp (e.g., current date and time) in configuration database **540** in association with unique AV identifier **502** to indicate when a last registration of AV **102** occurred.

[0078] In certain embodiments, configuration manager **534** notifies an operator of the registration, whereby the operator authorizes provisioning of AV **102**. If re-registration is required in edge cases (e.g., when AV certificate **504** has not automatically updated because AV **102** was offline for an extended period of time), an operator may interact with user interface **532** and configuration manager **534** to manually add a temporary exception to configuration definition **542** to allows registration and/or provisioning of AV **102** having the corresponding unique AV identifier **502**. Alternatively, to facilitate an initial registration of AV **102**, AV certificate **504** is manually installed onto AV **102**, where AV certificate **504** allows AV **102** to register with configuration manager **534**.

[0079] Advantageously, update of the software stack on AV **102** is simplified to changing configuration definition **542**, whereby software stack **510** is updated during a need boot of controller **406** of AV **102**. In certain embodiments, when configuration definition **542** is modified, configuration manager **534** sends a message to AV **102** to cause controller **406** to invoke AV configuration manager **508** to re-register and thereby update software stack **510**.

[0080] Configuration definitions **542** may have a hierarchical definitions where portions of configuration definition **542** that are common to multiple AVs **102** are shared. Advantageously, a fleet of AVs **102** at yard **104** are easily configured and/or updated in a more streamline way from VCM system **500**.

[0081] FIG. **6** is a flowchart illustrating one example VCM method **600**, in embodiments. VCM method **600** is for example implemented with configuration manager **534** of VCM server **530** and shows one example registration and provisioning of AV **102** by configuration manager **534**.

[0082] In block **610**, VCM method **600** interacts, via a user interface of a VCM server, with a user to receive a configuration definition of an AV. In one example of block **610**, configuration manager **534** interacts with an operator via user interface **532** to receive configuration definition **542**.

[0083] In block **615**, VCM method **600** stores the configuration definition in a configuration database in association with a unique AV identifier of the AV. In one example of block **615**, configuration manager **534** stores configuration definition **542** in configuration database **540**.

[0084] In block **620**, VCM method **600** associates the configuration definition with at least one credential stored and at least one software configuration in a credential vault, the at least one credential and the at least one software configuration enabling desired functionality to the AV. In one example of block **620**, configuration manager **534** associates configuration definition **542** with one or more software definitions **552** and/or credentials **554** of credential vault **550**.

[0085] In block **625**, VCM method **600** receives a registration request defining the unique identifier of the AV from an AV configuration manager of the AV. In one example of block **625**, configuration manager **534** receives registration request **560** from AV configuration manager **508** of AV **102**. In block **630**, VCM method **600** authenticates the registration request based on the unique AV identifier. In one example of block **630**, configuration manager **534** authenticates registration request **560** by verifying one or more of: the included unique AV identifier **502** is listed in configuration database **540**, the included AV certificate **504** is valid, the included current location **506** is within a defined operational area for yard **104**, and/or registration request **560** was received from an expected network (e.g., a network configured at yard **104**).

[0086] In block **635**, VCM method **600** retrieves the configuration definition from the configuration database based on the unique AV identifier. In one example of block **635**, configuration manager **534** retrieves configuration definition **542** from configuration database **540** based on unique AV identifier **502** received in registration request **560**.

[0087] In block **640**, VCM method **600** sends the configuration definition to the AV configuration manager in response to the registration request. In one example of block **640**, configuration manager **534** sends configuration status message **562** including configuration definition **542** to AV configuration manager **508** of AV **102**.

[0088] In block **645**, VCM method **600** receives, from the AV configuration manager, an update request identifying at least one of a requested software definition, and a requested credential. In one example of block **645**, configuration manager **534** received update request **564** from AV configuration manager **508** via AV API **536**.

[0089] In block **650**, VCM method **600** retrieves the at least one of the requested software definition, and the requested credential from the credential vault. In one example of block **650**, configuration manager **534** retrieves ones of software definition **552** and credentials **554** identified within update request **564**.

[0090] In block **655**, VCM method **600** sends the at least one of the requested software definition, and the requested credential to the AV configuration manager, wherein the AV configuration manager updates a software stack of the AV. In one example of block **655**, configuration manager **534** sends configuration message **566** including identified ones of software definition **552** and/or credentials **554** to AV configuration manager **508** of AV **102**.

[0091] FIG. **7** is a flowchart illustrating one example remote based control method **700** for autonomous operation, in embodiments. Method **700** is for example implemented in one or more of puller **122**, assessor **124**, scheduler **125**, and updater **127** of MC **120** of FIG. **1**.

[0092] In block **710**, method **700** pulls, from a yard management system to a mission control, a move request identifying a trailer, a starting parking spot, and a destination spot in a yard. In one example of block **710**, puller **122** pulls move request **108**(**3**) from move queue **110** of YMS **106**.

[0093] In block **720**, method **700** determines, at the mission control, whether the move request is feasible for an AV. In one example of block **720**, assessor **124** determines that move request **108**(**3**) meets criteria for an autonomous move by AV **102**. In another example of block **720**, assessor **124** determines that the move is not feasible by the AV when trailer **112**(**1**) is not parked within parking spot **114**(**1**), which is the starting parking spot.

[0094] In block **730**, method **700**, when the move request is feasible, generates a mission defining directives to control the AV to move the trailer from the starting parking spot to the destination spot. In one example of block **730**, scheduler **125** generates mission **128** with directives to control AV **102**(**1**) to hitch with trailer **112**(**1**) in parking spot **114**(**1**), drive trailer **112**(**1**) to, and reverse trailer **112**(**1**) into, parking spot **114**(**4**), and to unhitch from trailer **112**(**1**).

[0095] In block **740**, method **700** sets, via an API of the YMS, a status of the move request to indicate autonomous move scheduled. In one example of block **740**, updater **127** sets, via API **107**, status **109**(**3**) to indicate autonomous move scheduled.

[0096] FIG. **8** is a flowchart illustrating one example video bakery method **800** for an autonomous vehicle having a plurality of cameras, in embodiments. Video bakery method **800** is for example implemented by video bakery **330** within controller **406** of AV **102**.

[0097] In block **810**, method **800** receives a telemetry request identifying an area of interest and defining a maximum telemetry bandwidth. In one example of block **810**, bakery interface **452** receives telemetry request **326** from remote assistance application **442**.

[0098] In block **820**, method **800** selects at least two of the plurality of cameras based on a field of view for each camera and the area of interest. In one example of block **820**,

[0099] In block **830**, method **800** determines, based on the maximum telemetry bandwidth, a maximum frame rate and a maximum resolution of a single output video feed. In one example of block **830**,

[0100] In block **840**, method **800** processes video feeds of the at least two selected cameras to generate a still image of the area of interest. In one example of block **840**,

[0101] In block **850**, method **800** [processes the video feeds to generate the single output video feed to include at least part of each of the video feeds of the at least two selected cameras. In one example of block **850**,

[0102] In block **860**, method **800** outputs telemetry data including the single output video feed and the still image. In one example of block **860**,

[0103] Changes may be made in the above methods and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description or shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover all generic and specific features described herein, as well as all statements of the scope of the present method and system, which, as a matter of language, might be said to fall therebetween.

What is claimed is:

1. A remote based control method for autonomous operation, comprising:

receiving, at a mission control from a yard management system (YMS), a move request defining a destination spot in a yard and one or more of a trailer identifier of a trailer to be moved, a pick-up spot of the trailer, and a trailer type;

determining, at the mission control, whether the move request is feasible for an autonomous vehicle (AV); and

when the move request is feasible:

generating a mission defining directives to control the AV to move the trailer from the pick-up spot to the destination spot; and

setting, via an application programming interface (API) of the YMS, a status of the move request to indicate autonomous move scheduled.

2. The remote based control method of claim **1**, the determining comprising:

determining, based on weather data received at mission control, when operation of the AV is not possible;

updating an operational schema of the AV to indicate that operation of the AV is not possible; and

sending the operational schema to the YMS.

3. The remote based control method of claim **1**, the determining further comprising determining that the move request is not feasible when gladhand data indicates that the AV cannot move the trailer.

4. The remote based control method of claim **1**, the determining further comprising determining that the move request is not feasible when trailer information indicates that the trailer is damaged.

5. The remote based control method of claim **1**, setting the status of the move request within the YMS to indicate autonomous move not feasible when the move request is not feasible.

6. The remote based control method of claim **1**, when the move request defines the destination spot and one or both of the trailer identifier and the pick-up spot of the trailer, the determining comprising determining that (a) the trailer is parked in the pick-up spot, (b) the destination spot is empty, (c) at least one AV is available and capable of moving the trailer from the pick-up spot to the destination spot, and (d) the move meets certain criteria.

7. The remote based control method of claim **6**, the criteria comprising at least one of: the trailer being undamaged, there is no inventory issue for the trailer, and there are no issues at a loading dock when at least one of the pick-up spot and the destination spot is a loading dock.

8. The remote based control method of claim **7**, the issues at the loading dock including a dock light of the loading dock not being green.

9. The remote based control method of claim **1**, when the move request only defines the trailer identifier and the destination spot, the determining comprising searching an inventory of the YMS to determine the pick-up spot.

10. The remote based control method of claim **1**, when the move request only defines the trailer type and the destination spot, the determining comprising (a) searching a trailer database of the YMS to determine the trailer identifier of a trailer that matches the trailer type, and (b) searching an inventory of the YMS to determine the pick-up spot for the trailer having the trailer identifier.

11. The remote based control method of claim **1**, further comprising:

receiving, at the mission control, a mission status indicative of success from the AV when the mission is completed by the AV; and

updating the YMS to indicate that the move request is done.

12. The remote based control method of claim **11**, further comprising updating inventory information of the YMS to indicate that the pick-up spot is empty and that the destination spot contains the trailer having the trailer identifier.

13. The remote based control method of claim **1**, further comprising:

receiving a mission status from the AV indicating that the AV is unable to complete the mission; and

updating the YMS to indicate that the move request is not done.

14. The remote based control method of claim **1**, further comprising:

sending the mission to the AV;

receiving a mission status indicative of the destination spot not being empty from the AV;

searching an inventory of the YMS for an empty nearby spot; and

updating the mission to define directives to control the AV to deposit the trailer in the empty nearby spot.

15. The remote based control method of claim **1**, further comprising:

sending the mission to the AV;

receiving a mission status indicative of the destination spot not being empty from the AV;

determining, from an inventory of the YMS, that no nearby spot is empty;

sending an instruction to the AV to search for empty nearby spots;

receiving a message identifying an empty nearby spot from the AV; and

updating the mission to define directives to control the AV to deposit the trailer in the empty nearby spot.

16. The remote based control method of claim **1**, further comprising:

sending the mission to the AV;

receiving a mission status indicating that the trailer with the trailer identifier is not at the pick-up spot from the AV;

determining, from an inventory of the YMS, a new pick-up spot; and

updating the mission to define directives to control the AV to collect the trailer from the new pick-up spot.

17. The remote based control method of claim **1**, further comprising:

sending the mission to the AV;

receiving a mission status indicating that the trailer with the trailer identifier is not at the pick-up spot from the AV;

determining, from an inventory of the YMS, no new pick-up spot is found;

sending an instruction to the AV to identify nearby trailers;

receiving a message with a found trailer identifier from the AV;

determining that the found trailer identifier matches the trailer identifier; and

updating the mission to define directives to control the AV to collect the trailer from the new pick-up spot and move it to the destination spot.

* * * * *